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UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 20

SPECIAL CONTINOUS DISTRIBUTIONS

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter studies the three special continuous distributions of the A-Level syllabus — exponential, uniform (rectangular) and normal — together with their means, variances and probability computations. It ends with the use of the normal distribution, with continuity correction, as an approximation to the binomial and Poisson distributions, a favourite GCE examination topic.

- By the end of this chapter I can recognise a situation modelled by an exponential distribution and state its probability density function.
- By the end of this chapter I can compute probabilities, the mean $E(X)=1/\lambda$ and the variance $Var(X)=1/\lambda^2$ for an exponential distribution.
- By the end of this chapter I can define the uniform distribution on [a,b], and use its mean $(a+b)/2$ and variance $(b-a)^2/12$.
- By the end of this chapter I can state the properties of the normal distribution $N(\mu,\sigma^2)$ and interpret its density curve.
- By the end of this chapter I can standardise a normal variable using $Z=(X-\mu)/\sigma$ and read probabilities from the standard normal table $\Phi(z)$.
- By the end of this chapter I can solve inverse problems: find a value x or a parameter μ, σ given a probability.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. The Exponential Distribution
2. The Uniform (Rectangular) Distribution
3. The Normal Distribution: pdf, Mean and Variance
4. The Standard Normal Variable and Use of Tables
5. Normal Approximations to the Binomial and Poisson Distributions
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can recognise a situation modelled by an exponential distribution and state its probability density function.
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- By the end of this chapter I can solve inverse problems: find a value x or a parameter μ, σ given a probability.
- By the end of this chapter I can approximate a binomial distribution B(n,p) by a normal distribution when np and n(1−p) are large enough, applying the continuity correction.
- By the end of this chapter I can approximate a Poisson distribution $Po(\lambda)$ by $N(\lambda,\lambda)$ for large λ, applying the continuity correction.

PREREQUISITES

- Continuous random variables: probability density functions, $P(a\leq X\leq b)$ as an inte-

gral/area (Lower Sixth, Further Maths).

- Expectation E(X) and variance Var(X) of continuous random variables.
- The binomial distribution B(n,p) and the Poisson distribution Po(λ): conditions of use, means and variances.
- Integration of exponential functions ∫e^(−λx)dx and basic definite integrals.
- Reading statistical tables (as done for the binomial/Poisson tables in Lower Sixth).

1 THE EXPONENTIAL DISTRIBUTION

At the Mfoundi market in Yaoundé, a phone-repairer notices that the waiting time between two customers seems to follow a curious "memoryless" pattern, while the sacks of rice weighed by a wholesaler cluster around 50 kg with symmetric small errors. Meanwhile, an MTN engineer models the lifetime of a SIM card, and an examiner wants to know the chance that more than 60 of 100 candidates pass when each has probability 0.5 of passing. All these situations — waiting times, uniform errors, measurement fluctuations and large counts — are governed by **special continuous distributions**. This chapter gives you the essential tools of the GCE Advanced Level: the **exponential**, the **uniform** and the **normal** distributions, plus the powerful normal approximations to the binomial and Poisson laws.

We begin with a natural question: if events happen randomly in time at a steady average rate — customers arriving, phone calls received, machines breaking down — how long must we wait for the next event? The answer is the exponential distribution.

1.1 From Poisson events to waiting times

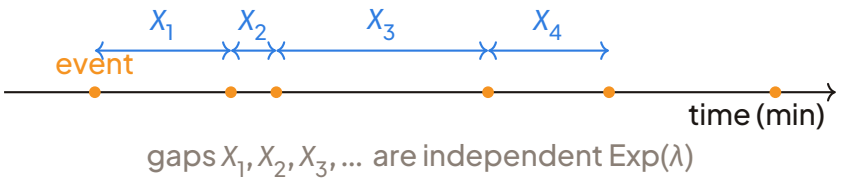
Recall the Poisson distribution: if events occur independently at a constant average rate of λ per unit time, then the number of events N in an interval of length t satisfies

$$P(N = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \qquad k = 0, 1, 2, \ldots$$

Now look at time differently. Instead of counting events, measure the waiting time X until the next event. The key observation is:

$$P(X > t) = P(\text{no event occurs in } [0, t]) = P(N = 0) = e^{-\lambda t}.$$

So the random waiting time X is a continuous random variable on $[0, +\infty)$ whose tail probability decays exponentially. Differentiating $P(X \le t) = 1 - e^{-\lambda t}$ with respect to t gives the density $\lambda e^{-\lambda t}$. This is exactly the model of the phone-repairer at Mfoundi: the gap between two customers is exponential.



Events of a Poisson process (orange dots) on a time axis; the waiting times between consecutive events are independent exponential random variables.

1.2 Definition and validity of the density

DEFINITION **Exponential distribution**

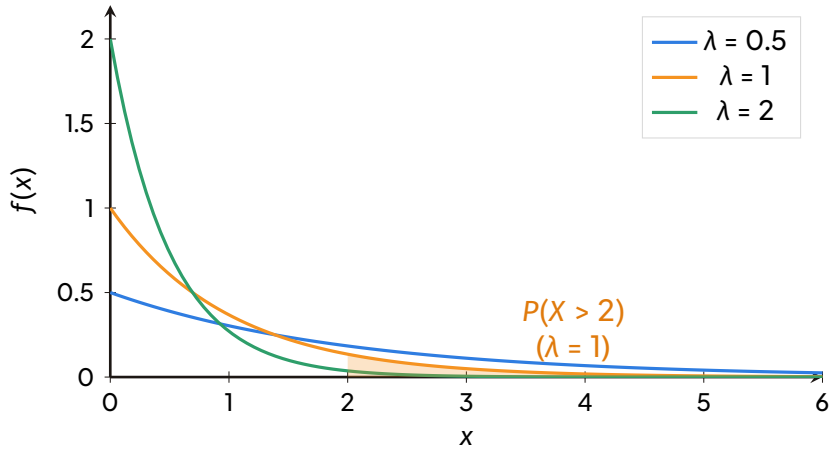
A continuous random variable X follows the **exponential distribution** with parameter $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$, if its probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Let us verify that f is a genuine pdf. First, since $\lambda > 0$ and $e^{-\lambda x} > 0$, we have $f(x) \geq 0$ for all x. Second, the total area under the curve is

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{0}^{+\infty} \lambda e^{-\lambda x}\,dx = \left[-e^{-\lambda x}\right]_0^{+\infty} = 0 - (-1) = 1.$$

Both conditions hold, so f is indeed a probability density function.



Densities of $\text{Exp}(\lambda)$ for $\lambda = 0.5, 1, 2$. Larger λ means faster decay and shorter typical waiting times. The shaded region is $P(X > 2) = e^{-2}$ for $\lambda = 1$.

1.3 Cumulative distribution function and tail probability

PROPERTY **cdf and tail probability**

If $X \sim \text{Exp}(\lambda)$, then for $x \geq 0$,

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \qquad P(X > x) = e^{-\lambda x}.$$

Proof. For $x \geq 0$,

$$F(x) = \int_0^x \lambda e^{-\lambda t}\,dt = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x},$$

and $P(X > x) = 1 - F(x) = e^{-\lambda x}$. □

METHOD **Computing $P(a \le X \le b)$**

For $0 \le a < b$ and $X \sim \text{Exp}(\lambda)$:

1. Write $P(a \le X \le b) = F(b) - F(a)$.
2. Substitute: $P(a \le X \le b) = (1 - e^{-\lambda b}) - (1 - e^{-\lambda a}) = e^{-\lambda a} - e^{-\lambda b}$.
3. Evaluate numerically, keeping at least 4 decimal places.

WARNING **Continuous variables: single points carry no probability**

Because X is continuous, $P(X = a) = 0$ for every a. Hence

$$P(a \le X \le b) = P(a < X < b) = P(a \le X < b) = P(a < X \le b).$$

Do not hesitate between strict and non-strict inequalities: for the exponential distribution they all give the same value.

1.4 Mean and variance

PROPERTY **Expectation and variance — proof examinable**

If $X \sim \text{Exp}(\lambda)$, then

$$E(X) = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Derivation. By definition, $E(X) = \int_0^{+\infty} x\,\lambda e^{-\lambda x}\,dx$. Integrate by parts with $u = x$, $dv = \lambda e^{-\lambda x}dx$, so $du = dx$, $v = -e^{-\lambda x}$:

$$E(X) = \left[-xe^{-\lambda x}\right]_0^{+\infty} + \int_0^{+\infty} e^{-\lambda x}\,dx = 0 + \left[-\frac{1}{\lambda}e^{-\lambda x}\right]_0^{+\infty} = \frac{1}{\lambda}.$$

(The bracket vanishes because $xe^{-\lambda x} \to 0$ as $x \to +\infty$.)

For the variance, compute $E(X^2) = \int_0^{+\infty} x^2\lambda e^{-\lambda x}\,dx$ by parts with $u = x^2$, $dv = \lambda e^{-\lambda x}dx$:

$$E(X^2) = \left[-x^2e^{-\lambda x}\right]_0^{+\infty} + 2\int_0^{+\infty} xe^{-\lambda x}\,dx = 0 + \frac{2}{\lambda}E(X) = \frac{2}{\lambda^2}.$$

Therefore

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \qquad \square$$

WARNING **λ is a rate, not a mean**

The parameter λ is a **rate** (events per unit time), not the mean. The mean waiting time is $E(X) = \frac{1}{\lambda}$. If the mean lifetime of a bulb is 500 hours, then $\lambda = \frac{1}{500} = 0.002$ per hour — not $\lambda = 500$. This confusion is the most frequent source of lost marks at the GCE. Note also that the standard deviation equals the mean: $\sigma = \frac{1}{\lambda} = E(X)$.

1.5 The memoryless property

PROPERTY **Memoryless property**

If $X \sim \text{Exp}(\lambda)$, then for all $s, t \geq 0$,

$$P(X > s + t \mid X > s) = P(X > t).$$

Proof. By the definition of conditional probability,

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)} = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t). \qquad \square$$

Interpretation. Suppose X is the lifetime of a component. Given that it has already survived s hours, the probability that it lasts at least t more hours is the same as the probability that a brand-new component lasts t hours. The component "forgets" its age: a used component is as good as new. This is exactly the pattern the Mfoundi phone-repairer observed: however long you have already waited for a customer, the remaining wait has the same distribution as at the start.

WORKED EXAMPLE 1 **Lifetime of an electronic component**

The lifetime X (in hours) of a certain electronic component follows an exponential distribution with mean 1000 hours.

(a) State the value of λ and write down the pdf of X.

(b) Find the probability that the component lasts more than 1200 hours.

(c) Find the probability that it lasts between 500 and 1500 hours.

(d) Given that it has already lasted 800 hours, find the probability that it lasts at least 400 more hours.

SOLUTION

Solution.

(a) Since $E(X) = \frac{1}{\lambda} = 1000$, we get $\lambda = \frac{1}{1000} = 0.001$ per hour. The pdf is

$$f(x) = 0.001\,e^{-0.001x}, \quad x \geq 0.$$

(b) $P(X > 1200) = e^{-0.001\times 1200} = e^{-1.2} \approx 0.3012$.

(c) $P(500 \leq X \leq 1500) = e^{-0.001\times 500} - e^{-0.001\times 1500} = e^{-0.5} - e^{-1.5} \approx 0.6065 - 0.2231 = 0.3834$.

(d) By the memoryless property,

$$P(X > 1200 \mid X > 800) = P(X > 400) = e^{-0.001\times 400} = e^{-0.4} \approx 0.6703.$$

WORKED EXAMPLE 2 **Finding the parameter from a probability**

The waiting time T (in minutes) between two arrivals at a bus stop in Bamenda is exponentially distributed. Given that $P(T > 10) = 0.6$, find:

(a) the parameter λ (to 4 decimal places);

(b) the mean waiting time;

(c) $P(T \leq 5)$.

SOLUTION

Solution.

(a) $P(T > 10) = e^{-10\lambda} = 0.6$, so $-10\lambda = \ln 0.6$ and

$$\lambda = -\frac{\ln 0.6}{10} = \frac{0.5108}{10} \approx 0.0511 \text{ per minute.}$$

(b) $E(T) = \frac{1}{\lambda} = \frac{10}{-\ln 0.6} \approx 19.58$ minutes.

(c) $P(T \leq 5) = 1 - e^{-5\lambda} = 1 - e^{-0.2554} \approx 1 - 0.7746 = 0.2254$.

KEY POINTS **Key points on the exponential distribution**

- $X \sim \text{Exp}(\lambda)$ models the **waiting time** between events of a Poisson process of rate λ; pdf $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- $F(x) = 1 - e^{-\lambda x}$ and $P(X > x) = e^{-\lambda x}$.
- $P(a \leq X \leq b) = e^{-\lambda a} - e^{-\lambda b}$.

- $E(X) = \frac{1}{\lambda}$, $\operatorname{Var}(X) = \frac{1}{\lambda^2}$, $\sigma = \frac{1}{\lambda}$.
- **Memoryless:** $P(X > s + t \mid X > s) = P(X > t)$.
- **Median** $m = \frac{\ln 2}{\lambda}$; **the mode is 0 (the density is decreasing).**

EXERCISE 1 **Calls at a call centre**

Calls arrive at a call centre in Douala at an average rate of 4 per minute, according to a Poisson process. Let T be the waiting time, in minutes, between two consecutive calls.

(a) Write down the distribution of T and its pdf.

(b) Calculate the probability of waiting more than 30 seconds between two calls.

(c) Calculate the probability of waiting between 10 and 20 seconds.

(d) Find the median waiting time, i.e. the value m such that $P(T > m) = 0.5$.

SOLUTION **Exercise 1**

(a) $T \sim \text{Exp}(4)$ with $f(t) = 4e^{-4t}$, $t \geq 0$ (t in minutes).

(b) 30 seconds = 0.5 min: $P(T > 0.5) = e^{-4\times 0.5} = e^{-2} \approx 0.1353$.

(c) 10 s = $\frac{1}{6}$ min, 20 s = $\frac{1}{3}$ min:

$$P\left(\tfrac{1}{6} \leq T \leq \tfrac{1}{3}\right) = e^{-4/6} - e^{-4/3} = e^{-2/3} - e^{-4/3} \approx 0.5134 - 0.2636 = 0.2498.$$

(d) $e^{-4m} = 0.5 \Rightarrow -4m = \ln 0.5 \Rightarrow m = \frac{\ln 2}{4} \approx 0.1733$ min ≈ 10.4 seconds.

EXERCISE 2 **Lifetime of a phone battery**

The lifetime X (in years) of a type of phone battery is modelled by $X \sim \text{Exp}(0.5)$.

(a) Find the mean and the standard deviation of X.

(b) Find the probability that a battery lasts more than 3 years.

(c) A battery has already worked for 2 years. Find the probability that it works for at least one more year.

(d) Find the value k such that $P(X \le k) = 0.9$.

SOLUTION **Exercise 2**

(a) $E(X) = \frac{1}{0.5} = 2$ years; $\sigma = \frac{1}{0.5} = 2$ years (and $\mathrm{Var}(X) = 4$).

(b) $P(X > 3) = e^{-0.5\times 3} = e^{-1.5} \approx 0.2231$.

(c) By the memoryless property, $P(X > 3 \mid X > 2) = P(X > 1) = e^{-0.5} \approx 0.6065$.

(d) $1 - e^{-0.5k} = 0.9 \Rightarrow e^{-0.5k} = 0.1 \Rightarrow k = \frac{\ln 10}{0.5} = 2\ln 10 \approx 4.61$ years.

DID YOU KNOW? **Why "memoryless" matters in engineering**

The exponential distribution is the only continuous distribution with the memoryless property. Engineers use it to model components whose failure is due to sudden external shocks rather than gradual wear — such as a fuse blown by a power surge. For components that age (tyres, batteries), other models are preferred, because an old tyre is definitely not as good as new!

2 THE UNIFORM (RECTANGULAR) DISTRIBUTION

In the previous section we met the **exponential distribution**, a continuous model whose probability density decreases as x increases: short waiting times are more likely than long ones. But not every continuous situation behaves that way. Imagine you arrive at a bus stop in Yaoundé knowing only that the bus will come at some moment within the next 15 minutes, with no reason to expect one moment more than another. Or think of a calculator that rounds every amount to the nearest franc: the rounding error is just as likely to be 0.1 franc as 0.4 franc. In such situations every value in an interval is equally likely. The model for this is the **uniform distribution**, the simplest of all continuous distributions, and the subject of this section.

2.1 Situations Leading to the Uniform Model

Before any formula, let us collect concrete situations where the uniform model is the natural choice.

- **A random number in an interval.** A computer (or a fair spinner) generates a number X "at random" between a and b. No sub-interval of a given length is favoured over another of the same length.

- **Rounding errors.** When a measurement is rounded to the nearest unit (nearest franc, nearest millimetre, nearest minute), the true value differs from the recorded value by an error X which lies uniformly between -0.5 and $+0.5$ of the unit.
- **Arrival within a known window.** A bus is known to arrive at some instant between 8:00 and 8:15, but nothing more precise is known. The arrival time X is uniform on $[0, 15]$ (measuring time in minutes after 8:00).
- **The position of a defect.** A flaw occurs at a uniformly random position along a cable of length L.

In each case the key phrase is: all values in the interval are equally likely. For a continuous variable this cannot mean that each single value has a positive probability (we know $P(X = x) = 0$ for continuous variables); it means that the **probability density function** is constant on the interval.

2.2 Definition and Probability Density Function

DEFINITION **The uniform distribution $U(a, b)$**

A continuous random variable X has the **uniform distribution** on the interval $[a, b]$ (with $a < b$), written $X \sim U(a, b)$, if its probability density function is

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

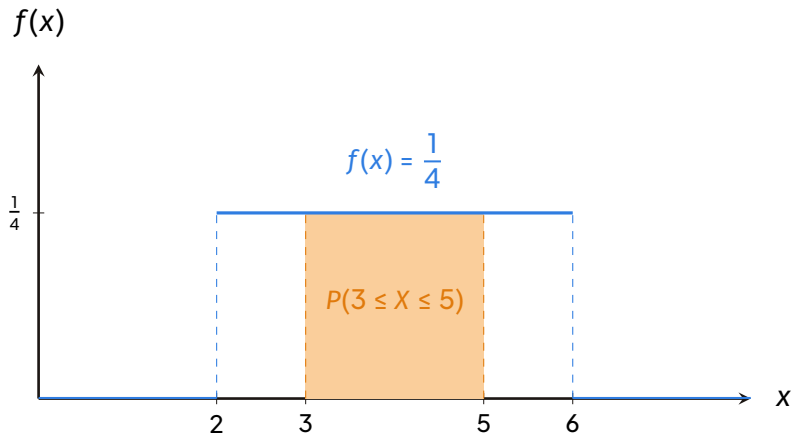
Because the graph of f is a horizontal segment over $[a, b]$, the region under the curve is a rectangle: this is why the distribution is also called the **rectangular distribution**.

Why must the constant height be exactly $\dfrac{1}{b-a}$? Because the total area under any pdf must equal 1. The region under the graph is a rectangle of base $b - a$ and height h, so

$$\text{Area} = (b-a) \times h = 1 \quad \Longrightarrow \quad h = \frac{1}{b-a}.$$

We can also verify this by integration:

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}\Big[x\Big]_a^b = \frac{b-a}{b-a} = 1.\checkmark$$



The pdf of $X \sim U(2,6)$. The shaded rectangle has area $P(3 \le X \le 5) = \frac{5-3}{6-2} = \frac{1}{2}$.

2.3 Probabilities as Ratios of Lengths

Since the density is constant, every probability is the area of a rectangle, and no integration is ever needed in practice.

PROPERTY **Probabilities for $U(a,b)$**

If $X \sim U(a,b)$ and $[c,d] \subseteq [a,b]$, then

$$P(c \le X \le d) = \int_c^d \frac{1}{b-a}\,dx = \frac{d-c}{b-a}.$$

In words: probability = (length of the favourable part) ÷ (total length). (The proof by integration, shown here, is examinable but immediate.)

METHOD **Computing a uniform probability**

1. Identify the interval $[a,b]$ and check that $X \sim U(a,b)$.
2. Identify the favourable interval $[c,d]$ and truncate it to $[a,b]$ if it sticks out.
3. Compute $P = \dfrac{\text{length of favourable part}}{b-a}$.

WARNING **Check the position of your bounds**

The formula $\frac{d-c}{b-a}$ is only valid when $[c,d] \subseteq [a,b]$. Remember that

$$P(X \le x) = 0 \text{ for } x < a, \qquad P(X \le x) = 1 \text{ for } x > b.$$

So, for example, if $X \sim U(2,6)$, then $P(X \le 5) = P(2 \le X \le 5) = \frac{3}{4}$, not $\frac{5}{4}$: the part of $(-\infty, 5]$ below $a = 2$ carries no probability at all. Always clip your interval to $[a,b]$ first.

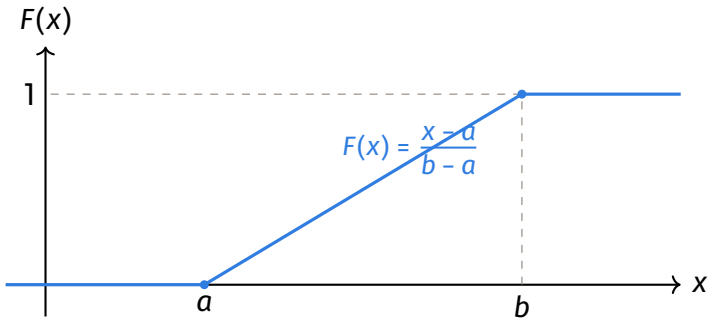
2.4 The Cumulative Distribution Function

Integrating the pdf from $-\infty$ to x gives the cumulative distribution function. For $x < a$ nothing has been accumulated; for $a \le x \le b$ the accumulated area is a rectangle of base $x - a$ and height $\frac{1}{b-a}$; for $x > b$ the whole area 1 is accumulated.

PROPERTY **CDF of $U(a,b)$**

If $X \sim U(a,b)$, its cumulative distribution function is

$$F(x) = P(X \le x) = \begin{cases} 0 & \text{if } x < a, \\ \frac{x-a}{b-a} & \text{if } a \le x \le b, \\ 1 & \text{if } x > b. \end{cases}$$



The cumulative distribution function of $U(a,b)$: flat at 0 before a, linear with slope $\frac{1}{b-a}$ on $[a,b]$, flat at 1 after b.

2.5 Mean and Variance of the Uniform Distribution

The mean. Intuitively, since the density is symmetric about the midpoint of $[a,b]$, the average

value must be the midpoint $\frac{a+b}{2}$. Let us prove it from the definition $E(X) = \int_{-\infty}^{+\infty} x f(x)\,dx$:

$$E(X) = \int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}.$$

The variance. We use the computational formula $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$. First,

$$E(X^2) = \int_a^b x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{b^3-a^3}{3(b-a)}.$$

Since $b^3 - a^3 = (b-a)(b^2+ab+a^2)$, this simplifies to $E(X^2) = \frac{a^2+ab+b^2}{3}$. Therefore

$$\mathrm{Var}(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$
$$= \frac{4(a^2+ab+b^2) - 3(a^2+2ab+b^2)}{12} = \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}.$$

PROPERTY **Mean and variance of $U(a,b)$**

If $X \sim U(a,b)$, then

$$E(X) = \frac{a+b}{2} \qquad \text{and} \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}, \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{b-a}{2\sqrt{3}}.$$

Both derivations above are examinable: know how to obtain them from $E(X) = \int x f(x)\,dx$ and $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

KEY POINTS **Key facts about $U(a,b)$**

- pdf: $f(x) = \frac{1}{b-a}$ on $[a,b]$, 0 elsewhere; the graph is a rectangle.
- Probabilities are ratios of lengths: $P(c \le X \le d) = \frac{d-c}{b-a}$ for $[c,d] \subseteq [a,b]$.
- $E(X) = \frac{a+b}{2}$ (the midpoint); $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$.
- The variance depends only on the width of the interval, not on its position.

2.6 Worked Examples

WORKED EXAMPLE 3 **Waiting for the bus**

A bus arrives at a stop at a time uniformly distributed between 8:00 and 8:15. Let X be the waiting time, in minutes after 8:00, so $X \sim U(0, 15)$.

1. Write down the pdf of X and find $E(X)$ and $\mathrm{Var}(X)$.
2. Find the probability of waiting between 3 and 9 minutes.
3. Find the probability of waiting more than 10 minutes.
4. Find the time t such that $P(X \le t) = 0.75$.

SOLUTION

Solution.

1. $f(x) = \dfrac{1}{15 - 0} = \dfrac{1}{15}$ for $0 \le x \le 15$, and 0 otherwise.

$$E(X) = \frac{0 + 15}{2} = 7.5 \text{ min}, \qquad \mathrm{Var}(X) = \frac{(15 - 0)^2}{12} = \frac{225}{12} = 18.75 \text{ min}^2.$$

2. $[3, 9] \subseteq [0, 15]$, so $P(3 \le X \le 9) = \dfrac{9 - 3}{15} = \dfrac{6}{15} = \dfrac{2}{5} = 0.4$.

3. $P(X > 10) = \dfrac{15 - 10}{15} = \dfrac{5}{15} = \dfrac{1}{3} \approx 0.333$.

4. We need $F(t) = \dfrac{t - 0}{15} = 0.75$, giving $t = 11.25$ minutes, i.e. 11 min 15 s after 8:00. There is a 75% chance the bus has arrived by then.

WORKED EXAMPLE 4 **Rounding to the nearest franc**

A shop in Douala rounds every bill to the nearest franc. The rounding error X (true amount minus rounded amount) is uniformly distributed on $[-0.5,\ 0.5]$ francs.

1. Find the mean and standard deviation of X.
2. Find the probability that the error exceeds 0.3 franc in absolute value.

SOLUTION

Solution. Here $a = -0.5$ and $b = 0.5$, so $b - a = 1$.

1. $E(X) = \dfrac{-0.5 + 0.5}{2} = 0$ franc: rounding is fair on average.

$$\mathrm{Var}(X) = \frac{1^2}{12} = \frac{1}{12}, \qquad \sigma = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}} \approx 0.289 \text{ franc}.$$

2. $P(|X| > 0.3) = P(X > 0.3) + P(X < -0.3) = \dfrac{0.5 - 0.3}{1} + \dfrac{-0.3 - (-0.5)}{1} = 0.2 + 0.2 = 0.4.$

2.7 Finding a and b from the Mean and Variance

In examination questions the endpoints are sometimes unknown and must be recovered from given information about the mean and the variance (or standard deviation). The strategy is to solve the two equations simultaneously.

METHOD **Recovering a and b from $E(X)$ and $\mathrm{Var}(X)$**

1. Write $\dfrac{a+b}{2} = \mu$ and $\dfrac{(b-a)^2}{12} = \sigma^2$.
2. From the second equation, $b - a = \sqrt{12\sigma^2} = 2\sqrt{3}\,\sigma$ (take the positive root since $b > a$).
3. Solve the system $a + b = 2\mu$, $b - a = 2\sqrt{3}\,\sigma$ by adding and subtracting:

$$a = \mu - \sqrt{3}\,\sigma, \qquad b = \mu + \sqrt{3}\,\sigma.$$

WORKED EXAMPLE 5 **Finding the endpoints**

A continuous random variable X is uniformly distributed on $[a, b]$. Given that $E(X) = 4$ and $\mathrm{Var}(X) = 3$, find a and b, and hence compute $P(2 \le X \le 5)$.

SOLUTION

Solution. We set up the two equations:

$$\frac{a+b}{2} = 4 \implies a + b = 8; \qquad \frac{(b-a)^2}{12} = 3 \implies (b-a)^2 = 36 \implies b - a = 6$$

(taking the positive square root because $b > a$). Adding the equations: $2b = 14$, so $b = 7$; subtracting: $2a = 2$, so $a = 1$. Hence $X \sim U(1, 7)$ and

$$P(2 \le X \le 5) = \frac{5 - 2}{7 - 1} = \frac{3}{6} = \frac{1}{2}.$$

WARNING **Trap: the sign of the square root**

From $(b - a)^2 = 36$ you must take $b - a = +6$, not -6, because by definition $a < b$. Choosing the negative root swaps a and b and gives an invalid interval. Also, do not confuse $\mathrm{Var}(X)$

with σ: if the question gives the standard deviation, square it before using $\dfrac{(b-a)^2}{12}$.

EXERCISE 3 **Practice**

1. $X \sim U(3, 11)$. Find (a) $E(X)$, (b) $\mathrm{Var}(X)$, (c) $P(X \le 5)$, (d) $P(X > 13)$, (e) the value m such that $P(X \le m) = \frac{1}{4}$.
2. The error made when a stopwatch reading is rounded to the nearest second is uniform on $[-0.5, 0.5]$ seconds. Find the probability that the error is within 0.2 second of zero.
3. $X \sim U(a, b)$ with $E(X) = 10$ and $\sigma = 2\sqrt{3}$. Find a and b.

SOLUTION **Exercise 1**

1. (a) $E(X) = \dfrac{3+11}{2} = 7$. (b) $\mathrm{Var}(X) = \dfrac{(11-3)^2}{12} = \dfrac{64}{12} = \dfrac{16}{3}$. (c) $P(X \le 5) = \dfrac{5-3}{8} = \dfrac{1}{4}$. (d) $P(X > 13) = 0$ since $13 > b = 11$. (e) $\dfrac{m-3}{8} = \dfrac{1}{4} \Rightarrow m - 3 = 2 \Rightarrow m = 5$.
2. $P(|X| \le 0.2) = P(-0.2 \le X \le 0.2) = \dfrac{0.2 - (-0.2)}{1} = 0.4$.
3. $b - a = 2\sqrt{3}\,\sigma = 2\sqrt{3} \times 2\sqrt{3} = 12$ and $a + b = 20$. Adding: $2b = 32$, $b = 16$; then $a = 4$. So $X \sim U(4, 16)$.

DID YOU KNOW? **Why "rectangular"?**

The uniform distribution is the continuous analogue of throwing a fair die: a fair die gives equal probability $\frac{1}{6}$ to each of six discrete outcomes, while $U(a, b)$ spreads the probability evenly over a whole continuum of values. Random-number generators in calculators and computers produce (approximately) $U(0, 1)$ values, and every other continuous distribution used in simulation — including the exponential distribution of the previous section and the normal distribution of the next — is built by transforming these uniform numbers.

The uniform distribution models complete indifference within an interval. But most real measurements — heights, marks, masses — cluster around a central value and become rarer as we move away from it. Modelling that bell-shaped behaviour is the work of the normal distribution, which we study next.

3 THE NORMAL DISTRIBUTION: PDF, MEAN AND VARIANCE

3.1 From Uniform to Normal: Why the Bell Curve?

In the previous section we studied the uniform (rectangular) distribution, where every value in an interval is equally likely. Many natural phenomena, however, do not behave like that. Consider the heights of Upper Sixth students in a Cameroonian secondary school, the marks obtained in a well-designed Further Mathematics examination, or the errors made when measuring the length of a laboratory table with a ruler. In each case:

- Most observations cluster around a central value.
- Values far from the centre become progressively rarer.
- The pattern is symmetric: a student is roughly as likely to be 5 cm above the mean height as 5 cm below it.

This symmetric, bell-shaped pattern is so ubiquitous that it has a name: the **normal distribution** (also called the **Gaussian distribution**). A deep mathematical result — the Central Limit Theorem — tells us that the sum (or average) of a large number of small, independent random effects tends to follow this distribution, regardless of the original distributions of those effects. This explains why the normal distribution appears everywhere: heights, measurement errors, examination marks, lifetimes of manufactured items (under certain conditions), and many more.

3.2 Definition and Probability Density Function

DEFINITION **Normal Distribution $N(\mu, \sigma^2)$**

A continuous random variable X follows a normal distribution with mean μ and variance σ^2 (where $\mu \in \mathbb{R}$ and $\sigma > 0$) if its probability density function (pdf) is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad x \in \mathbb{R}.$$

We write $X \sim N(\mu, \sigma^2)$.

WARNING **Variance, not Standard Deviation!**

The second parameter in the notation $N(\mu, \sigma^2)$ is the variance σ^2, not the standard deviation σ. This is a very common examination error. Always check: if a question says $X \sim N(10, 4)$, then $\mu = 10$ and $\sigma^2 = 4$, so $\sigma = 2$.

The constants in the pdf have precise roles:

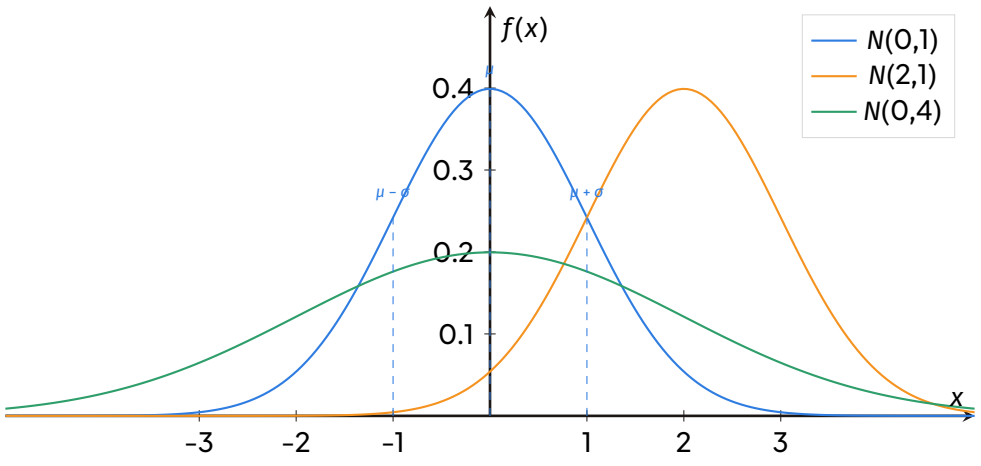
- $\frac{1}{\sigma\sqrt{2\pi}}$ is the normalising constant that makes the total area under the curve equal to 1.
- The exponent $-\frac{(x-\mu)^2}{2\sigma^2}$ is a quadratic in x, centred at μ and scaled by σ. It guarantees the bell shape and the symmetry.

3.3 Shape and Properties of the Normal Curve

PROPERTY **Key Features of the Normal Curve**

For $X \sim N(\mu, \sigma^2)$ with pdf $f(x)$:

1. Bell-shaped and symmetric about the vertical line $x = \mu$. Hence $f(\mu + d) = f(\mu - d)$ for any d.
2. Maximum at $x = \mu$; the maximum value is $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$.
3. Points of inflection at $x = \mu - \sigma$ and $x = \mu + \sigma$. The curve changes from concave down (near the centre) to concave up (in the tails) exactly one standard deviation away from the mean.
4. Asymptotic to the x-axis: $f(x) \to 0$ as $x \to \pm\infty$, but never touches the axis.
5. Total area under the curve is 1: $\int_{-\infty}^{\infty} f(x)\,dx = 1$.



Normal curves: $N(0,1)$ (blue), $N(2,1)$ (orange), $N(0,4)$ (green). The blue curve shows inflection points at $\mu \pm \sigma$.

3.4 Role of the Parameters μ and σ

The figure above illustrates perfectly how the two parameters control the distribution:

- **μ (location)**: shifts the curve left or right without changing its shape. $N(2,1)$ is simply $N(0,1)$

moved two units to the right.

- σ **(spread)**: stretches or compresses the curve horizontally. $N(0, 4)$ has $\sigma = 2$; it is wider and lower than $N(0, 1)$ (which has $\sigma = 1$). The area remains 1, so a wider curve must be lower at the peak.

KEY POINTS **Interpreting μ and σ**

- μ is the centre of the distribution (mean, median, and mode all coincide).
- σ is the standard deviation; it measures the typical distance of values from the mean. A larger σ means more variability, a flatter and wider bell.
- The inflection points are always exactly at $\mu \pm \sigma$. This gives a visual way to estimate σ from a sketch.

3.5 Expectation and Variance

PROPERTY **Mean and Variance of the Normal Distribution**

If $X \sim N(\mu, \sigma^2)$, then

$$\mathbb{E}(X) = \mu, \qquad \mathrm{Var}(X) = \sigma^2, \qquad \sigma_X = \sigma.$$

EXPERIMENT **Derivation of $\mathbb{E}(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$ (for insight, not examinable as a full proof)**

We compute $\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x)\,dx$. Substitute $z = \frac{x-\mu}{\sigma}$, so $x = \mu + \sigma z$, $dx = \sigma\,dz$. Then

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} (\mu + \sigma z)\frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz = \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz}_{=1} + \sigma \underbrace{\int_{-\infty}^{\infty} z\frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz}_{=0 \text{ (odd function)}} = \mu.$$

For the variance, $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$. With the same substitution $z = \frac{x-\mu}{\sigma}$, $(x-\mu)^2 = \sigma^2 z^2$, $dx = \sigma\,dz$:

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} \sigma^2 z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz = \sigma^2 \int_{-\infty}^{\infty} z^2 \phi(z)\,dz \quad \text{where } \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Integrate by parts: let $u = z$, $dv = z\phi(z)\,dz$. Then $du = dz$, $v = -\phi(z)$ (since $\frac{d}{dz}\phi(z) = -z\phi(z)$).

$$\int_{-\infty}^{\infty} z^2 \phi(z)\,dz = \Big[-z\phi(z)\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z)\,dz = 0 + 1 = 1.$$

Hence $\mathrm{Var}(X) = \sigma^2$.

3.6 Probabilities as Areas Under the Curve

For any continuous distribution, $P(a \le X \le b)$ is the area under the pdf curve between a and b. For the normal distribution, symmetry gives immediate results:

PROPERTY **Symmetry Probabilities**

If $X \sim N(\mu, \sigma^2)$, then

$$P(X \le \mu) = P(X \ge \mu) = 0.5.$$

More generally, for any $d > 0$,

$$P(\mu - d \le X \le \mu + d) = 2\,P(\mu \le X \le \mu + d).$$

METHOD **Sketching Before Calculating**

1. Draw a bell-shaped curve centred at μ.
2. Mark the values a and b (or the relevant tail) on the x-axis.
3. Shade the required area.
4. Use symmetry to relate the shaded area to standard normal probabilities (next section).

3.7 The Empirical Rule (68–95–99.7 Rule)

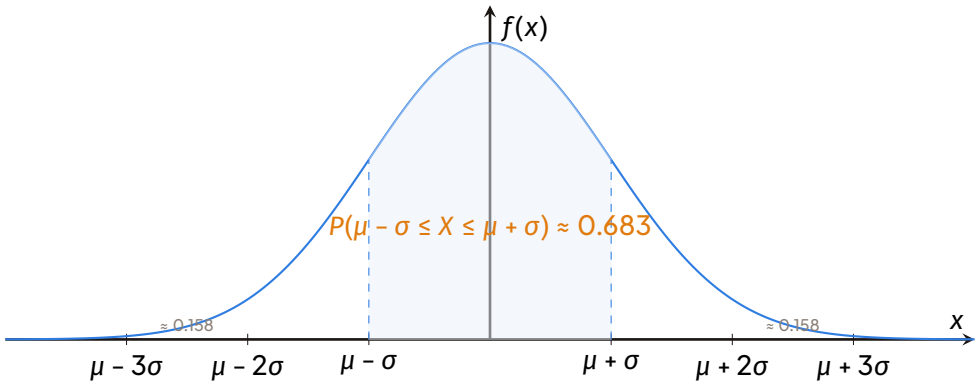
Although we cannot integrate the normal pdf in closed form (there is no elementary antiderivative), the areas within 1, 2, and 3 standard deviations of the mean are fixed constants that every student should know. They provide an excellent intuitive grasp of what σ means.

PROPERTY **Benchmark Areas — The Empirical Rule**

For $X \sim N(\mu, \sigma^2)$:

$$\begin{aligned} P(\mu - \sigma \le X \le \mu + \sigma) &\approx 0.6827 \quad \text{(about 68.3\%)},\\ P(\mu - 2\sigma \le X \le \mu + 2\sigma) &\approx 0.9545 \quad \text{(about 95.4\%)},\\ P(\mu - 3\sigma \le X \le \mu + 3\sigma) &\approx 0.9973 \quad \text{(about 99.7\%)}. \end{aligned}$$

Equivalently, the tail probabilities are: $P(|X - \mu| > \sigma) \approx 0.317$, $P(|X - \mu| > 2\sigma) \approx 0.045$, $P(|X - \mu| > 3\sigma) \approx 0.0027$.



The central 68.3% of the area lies within one standard deviation of the mean.

3.8 Why We Need Standardisation — Bridge to the Next Section

The integrals $\int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,dx$ cannot be expressed in terms of elementary functions. There is no "normal antiderivative" like $\frac{x^2}{2}$ or $\ln x$. This is why we **standardise** every normal variable to the **standard normal variable** $Z \sim N(0,1)$ and use pre-computed tables (or calculators). The next section (Lesson 142) will show exactly how to do this.

WORKED EXAMPLE 6 **Interpreting Parameters and Empirical Rule**

The weights of bags of rice produced by a factory in Douala are normally distributed with mean 50.2 kg and standard deviation 0.4 kg.

1. Write down the distribution of X, the weight of a randomly chosen bag.
2. Between what two weights do approximately 95% of the bags lie?
3. A bag is selected at random. Without using tables, estimate the probability that it weighs more than 51.0 kg.

SOLUTION

Solution.

1. Mean $\mu = 50.2$, standard deviation $\sigma = 0.4$, so variance $\sigma^2 = 0.16$.

$$X \sim N(50.2,\ 0.16).$$

2. By the empirical rule, 95% of the data lie within 2σ of the mean.

$$\mu \pm 2\sigma = 50.2 \pm 2(0.4) = 50.2 \pm 0.8.$$

So approximately 95% of bags weigh between 49.4 kg and 51.0 kg.

3. 51.0 kg is exactly $\mu + 2\sigma$. The area to the right of $\mu + 2\sigma$ is half of the outer 4.55% (since 95.45% is inside).
$$P(X > 51.0) \approx \frac{1 - 0.9545}{2} = 0.02275 \approx 2.3\%.$$
(Using the more precise 95.45% figure gives 2.275%; using 95% gives 2.5%. Both are acceptable estimates without tables.)

3.9 Common Mistakes and Exam Tips

WARNING **Frequent Errors**

- Writing $N(\mu, \sigma)$ instead of $N(\mu, \sigma^2)$. Always square the standard deviation to get the variance for the notation.
- Forgetting that the normal distribution is continuous: $P(X = a) = 0$ for any specific value a. Only intervals have non-zero probability.
- Confusing the pdf $f(x)$ (which can be > 1 for small σ) with a probability. $f(x)$ is a density, not a probability.
- Trying to "integrate the normal pdf" in an exam. You cannot. You must standardise and use tables.

KEY POINTS **Exam Checklist for This Section**

- State the pdf correctly: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$.
- Identify μ and σ^2 from $N(\mu, \sigma^2)$ notation.
- Sketch the curve: label μ, inflection points at $\mu \pm \sigma$, show symmetry.
- Recall $\mathbb{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$.
- Know the empirical rule percentages (68.3%, 95.4%, 99.7%) for quick estimation and checking.
- Always draw a diagram and shade the required area before reaching for the tables.

EXERCISE 4 **Practice**

1. A random variable $X \sim N(15, 9)$. Write down the mean, variance, and standard devia-

tion. Sketch the pdf, marking the mean and the inflection points.

2. The lifetimes of a certain brand of LED bulb are normally distributed with mean 12000 hours and variance 10^6 hours^2. Use the empirical rule to estimate the percentage of bulbs that last between 10000 and 14000 hours.
3. If $Y \sim N(\mu, 25)$ and $P(Y > 30) = 0.5$, find μ.
4. Explain why the total area under any normal curve is exactly 1, referring to the constant $\frac{1}{\sigma\sqrt{2\pi}}$.

SOLUTION **Exercise 1**

1. $\mu = 15$, $\sigma^2 = 9$, $\sigma = 3$. Inflection points at 15 ± 3, i.e. 12 and 18. The curve is symmetric about $x = 15$, peak height $f(15) = \frac{1}{3\sqrt{2\pi}} \approx 0.133$.
2. $\mu = 12000$, $\sigma = \sqrt{10^6} = 1000$. The interval $[10000, 14000]$ is $\mu \pm 2\sigma$. By the empirical rule, $\approx 95.4\%$ (or 95%).
3. $P(Y > 30) = 0.5$ means 30 is the median. For a normal distribution, median $= \mu$. Hence $\mu = 30$.
4. The constant $\frac{1}{\sigma\sqrt{2\pi}}$ is chosen precisely so that $\int_{-\infty}^{\infty} f(x)\,dx = 1$. This is a defining property of any probability density function.

4 THE STANDARD NORMAL VARIABLE AND USE OF TABLES

In the previous section we discovered the normal distribution $N(\mu, \sigma^2)$, with its bell-shaped density curve centred on the mean μ and spread out by the standard deviation σ. We also saw that probabilities are areas under the curve. But here is the practical difficulty: the density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

has no elementary antiderivative. We cannot integrate it by hand. So how did statisticians compute normal probabilities before computers? The answer is one of the cleverest ideas in all of statistics: instead of building a table for every possible μ and σ (there are infinitely many!), we build one single table for one special normal distribution, and we transform every other normal variable into that special one. That special distribution is the **standard normal distribution**, and the transformation is called **standardisation**. This section is the computational heart of the whole chapter: nearly every GCE question on the normal distribution reduces to the techniques

you are about to learn.

4.1 The Standard Normal Distribution

DEFINITION **Standard normal distribution and the function Φ**

The **standard normal distribution** is the normal distribution with mean 0 and variance 1. A variable following it is usually denoted Z, and we write

$$Z \sim N(0, 1).$$

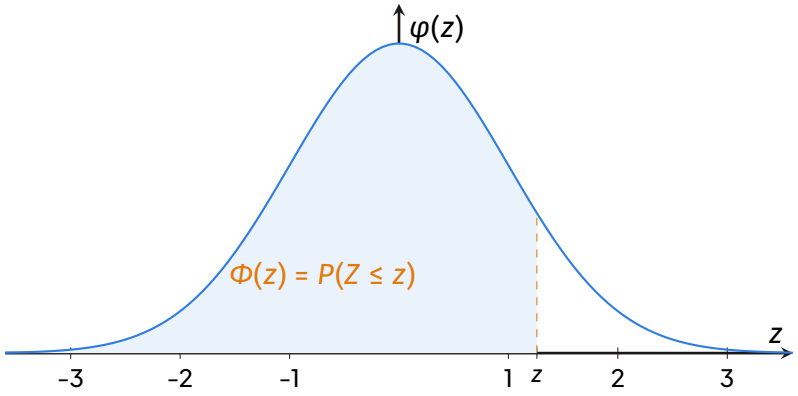
Its probability density function is

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \qquad z \in \mathbb{R}.$$

Its cumulative distribution function is denoted by the capital Greek letter **Phi**:

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\, dt.$$

Geometrically, $\Phi(z)$ is simply the area under the standard normal curve to the left of z. Since the total area under the curve is 1, we always have $0 < \Phi(z) < 1$, with $\Phi(z) \to 0$ as $z \to -\infty$ and $\Phi(z) \to 1$ as $z \to +\infty$.



The standard normal curve. The shaded left tail is the cumulative probability $\Phi(z) = P(Z \le z)$.

PROPERTY **Elementary properties of Φ**

For the standard normal variable $Z \sim N(0, 1)$:

1. $\Phi(0) = 0.5$ (the curve is symmetric about 0, so half the area lies to the left of 0);

2. $\Phi(+\infty) = 1$ and $\Phi(-\infty) = 0$;
3. $P(Z = z) = 0$ for every single value z (a continuous variable has zero probability of hitting an exact point), so $P(Z \le z) = P(Z < z)$ and $P(Z \ge z) = P(Z > z)$.

4.2 The Standardisation Theorem

Why is $N(0,1)$ enough for everything? Because of the following fundamental result.

PROPERTY **Standardisation theorem — proof instructive**

If $X \sim N(\mu, \sigma^2)$, then the variable

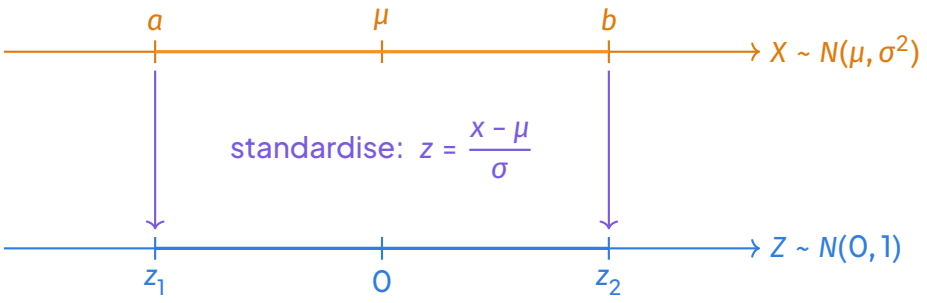
$$Z = \frac{X - \mu}{\sigma} \quad \text{satisfies} \quad Z \sim N(0,1).$$

Conversely, $X = \mu + \sigma Z$.

Justification. Subtracting μ shifts the centre of the distribution from μ to 0; dividing by σ rescales the horizontal axis so that the spread becomes 1. More formally, a linear transformation $aX + b$ of a normal variable is still normal, with $E(aX + b) = a\mu + b$ and $\text{Var}(aX + b) = a^2\sigma^2$. Taking $a = \frac{1}{\sigma}$ and $b = -\frac{\mu}{\sigma}$:

$$E(Z) = \frac{1}{\sigma}\mu - \frac{\mu}{\sigma} = 0, \qquad \text{Var}(Z) = \frac{1}{\sigma^2} \cdot \sigma^2 = 1.$$

Hence Z is normal with mean 0 and variance 1, i.e. $Z \sim N(0,1)$. ■



Standardisation maps the interval $[a, b]$ under X to the interval $[z_1, z_2]$ under Z, where $z_1 = \frac{a-\mu}{\sigma}$ and $z_2 = \frac{b-\mu}{\sigma}$. The two intervals carry the same probability.

The practical consequence is enormous: every probability question about any normal variable X becomes a question about Z, whose cumulative probabilities are tabulated.

METHOD **Computing normal probabilities by standardisation**

1. Identify μ and σ (take the square root if the variance is given).
2. Rewrite the event in terms of X, then standardise each bound: replace x by $z = \dfrac{x-\mu}{\sigma}$.
3. Use the table to find $\Phi(z)$:
   - $P(X \le x) = \Phi\left(\dfrac{x-\mu}{\sigma}\right)$;
   - $P(X \ge x) = 1 - \Phi\left(\dfrac{x-\mu}{\sigma}\right)$;
   - $P(a \le X \le b) = \Phi\left(\dfrac{b-\mu}{\sigma}\right) - \Phi\left(\dfrac{a-\mu}{\sigma}\right)$.
4. Round the final probability to 4 decimal places, as in the table.

4.3 Reading the Standard Normal Table

The GCE standard normal table gives values of $\Phi(z)$ for $z \ge 0$, to **two decimal places** in z and **four decimal places** in $\Phi(z)$. The table is organised as follows:

- the **row** gives z to one decimal place (e.g. 1.2);
- the **column** gives the second decimal digit (e.g. 0.06);
- the entry at the intersection is $\Phi(z)$.

EXEMPLE **Reading $\Phi(1.26)$**

To find $\Phi(1.26)$: go down the left column to the row 1.2, then across to the column headed 0.06 (the second decimal digit). The entry is 0.8962. Hence

$$\Phi(1.26) = P(Z \le 1.26) = 0.8962.$$

z	0.04	0.05	0.06	0.07	0.08
1.1	0.8729	0.8749	0.8770	0.8790	0.8810
1.2	0.8925	0.8944	**0.8962**	0.8980	0.8997
1.3	0.9099	0.9115	0.9131	0.9147	0.9162

Extract-style sketch of the standard normal table. Row 1.2, column 0.06 gives $\Phi(1.26) = 0.8962$.

4.4 Symmetry Relations: Dealing with Negative z and Upper Tails

The table only lists $z \geq 0$. For negative values and for upper-tail probabilities we exploit the symmetry of the curve about 0.

PROPERTY **Symmetry relations**

For any real numbers z, a, b with $a \leq b$:

$$\Phi(-z) = 1 - \Phi(z), \qquad P(Z > z) = 1 - \Phi(z), \qquad P(a \leq Z \leq b) = \Phi(b) - \Phi(a).$$

Why $\Phi(-z) = 1 - \Phi(z)$? By symmetry of the curve about the vertical axis, the area to the left of $-z$ equals the area to the right of z:

$$\Phi(-z) = P(Z \leq -z) = P(Z \geq z) = 1 - P(Z < z) = 1 - \Phi(z).$$

WARNING **The most frequent error with Φ**

Never write $\Phi(-z) = -\Phi(z)$. Probabilities cannot be negative! The correct relation is

$$\Phi(-z) = 1 - \Phi(z).$$

For example $\Phi(-1.26) = 1 - \Phi(1.26) = 1 - 0.8962 = 0.1038$, not -0.8962.

WORKED EXAMPLE 7 **Direct probability calculations for Z**

Given $Z \sim N(0,1)$, find (a) $P(Z \leq 1.5)$; (b) $P(Z > 0.84)$; (c) $P(Z \leq -2)$; (d) $P(-1.5 \leq Z \leq 2)$; (e) $P(|Z| \leq 1.96)$.

SOLUTION

Solution.

1. $P(Z \leq 1.5) = \Phi(1.5) = 0.9332$.
2. $P(Z > 0.84) = 1 - \Phi(0.84) = 1 - 0.7995 = 0.2005$.
3. $P(Z \leq -2) = \Phi(-2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$.
4. $P(-1.5 \leq Z \leq 2) = \Phi(2) - \Phi(-1.5) = \Phi(2) - \big(1 - \Phi(1.5)\big) = 0.9772 - (1 - 0.9332) = 0.9104$.
5. $P(|Z| \leq 1.96) = P(-1.96 \leq Z \leq 1.96) = \Phi(1.96) - \Phi(-1.96) = 2\Phi(1.96) - 1 = 2(0.9750) - 1 = 0.9500$.

Note the useful shortcut from part (e): $P(|Z| \leq z) = 2\Phi(z) - 1$. This is why 95% of a normal distribution lies within 1.96 standard deviations of the mean.

4.5 Worked Examples with Real Data

WORKED EXAMPLE 8 **Marks in a mock examination**

The marks obtained by Upper Sixth students in a Further Mathematics mock examination in Yaoundé are modelled by $X \sim N(58, 12^2)$. A student is chosen at random. Find the probability that the student scored (a) at most 70 marks; (b) at least 40 marks; (c) between 50 and 70 marks.

SOLUTION

Solution. Here $\mu = 58$ and $\sigma = 12$. Standardise with $Z = \dfrac{X - 58}{12}$.

1. $P(X \le 70) = P\left(Z \le \dfrac{70 - 58}{12}\right) = P(Z \le 1) = \Phi(1) = 0.8413.$

2. $P(X \ge 40) = P\left(Z \ge \dfrac{40 - 58}{12}\right) = P(Z \ge -1.5) = P(Z \le 1.5) = \Phi(1.5) = 0.9332.$

3. $P(50 \le X \le 70) = P\left(\dfrac{50 - 58}{12} \le Z \le \dfrac{70 - 58}{12}\right) = P(-0.67 \le Z \le 1)$

$$= \Phi(1) - \Phi(-0.67) = \Phi(1) - \big(1 - \Phi(0.67)\big) = 0.8413 - (1 - 0.7486) = 0.5899.$$

So about 84% of students scored at most 70, about 93% scored at least 40, and about 59% scored between 50 and 70.

WORKED EXAMPLE 9 **Masses of bags of cement**

The mass of a bag of cement produced by a factory in Douala is normally distributed with mean 50 kg and standard deviation 0.4 kg. A bag is rejected if its mass is below 49.2 kg. Find the proportion of bags rejected.

SOLUTION

Solution. Let $M \sim N(50, 0.4^2)$. Standardise the bound 49.2:

$$z = \frac{49.2 - 50}{0.4} = -2.$$

Hence

$$P(M < 49.2) = P(Z < -2) = \Phi(-2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228.$$

About 2.28% of the bags are rejected.

4.6 Inverse Problems of Type 1: Finding a Value from a Probability

So far we were given a value x and asked for a probability. Inverse problems reverse the ques-

tion: we are given the probability p and asked for the value x such that $P(X \le x) = p$.

METHOD **Inverse problem: from probability to value**

1. Write the statement in the form $P(X \le x) = p$ (use $1 - \Phi$ first if the given event is of the type $P(X \ge x)$).
2. Read the table backwards: look for p (or the closest entry) in the body of the table and read off z from the row and column headings. If $p < 0.5$, then z is negative: find z' with $\Phi(z') = 1 - p$ and set $z = -z'$.
3. Un-standardise: $x = \mu + \sigma z$.

WORKED EXAMPLE 10 **Finding a diameter exceeded by 5% of items**

The diameters of ball-bearings made by a machine are distributed as $D \sim N(25, 0.04^2)$ mm. Find the diameter d exceeded by only 5% of the ball-bearings.

SOLUTION

Solution. "Exceeded by 5%" means $P(D > d) = 0.05$, i.e. $P(D \le d) = 0.95$. Standardising,

$$P\left(Z \le \frac{d - 25}{0.04}\right) = 0.95 \quad \Longrightarrow \quad \Phi\left(\frac{d - 25}{0.04}\right) = 0.95.$$

Reading the table backwards, the entry closest to 0.9500 corresponds to $z = 1.645$ (mid-way between 1.64 giving 0.9495 and 1.65 giving 0.9505). Therefore

$$\frac{d - 25}{0.04} = 1.645 \quad \Longrightarrow \quad d = 25 + 0.04(1.645) = 25.0658 \approx 25.07 \text{ mm}.$$

Only 5% of the ball-bearings have a diameter greater than about 25.07 mm.

4.7 Inverse Problems of Type 2: Finding an Unknown μ or σ

Sometimes the mean or the standard deviation (or both) is unknown, but we are given one or two probability statements. Each probability statement, after standardisation, produces one equation involving μ and σ.

WARNING **Two probabilities ⇒ two equations**

When both μ and σ are unknown, a single probability statement is not enough: one equation cannot determine two unknowns. You need two probability statements, giving two simultaneous equations in μ and σ. Solve them by elimination or substitution.

WORKED EXAMPLE 11 **Determining μ and σ from two probabilities**

The masses of loaves of bread sold by a bakery are normally distributed with unknown mean μ grams and unknown standard deviation σ grams. It is known that 10% of the loaves have mass less than 780 g and 20% have mass greater than 830 g. Find μ and σ.

SOLUTION

Solution. Translate each statement into an equation in Z.

$$P(X < 780) = 0.10 \implies \Phi\left(\frac{780 - \mu}{\sigma}\right) = 0.10.$$

Since $0.10 < 0.5$, the z-value is negative. From the table, $\Phi(1.2816) \approx 0.90$, so $\Phi(-1.2816) = 0.10$. Hence

$$\frac{780 - \mu}{\sigma} = -1.2816 \quad \implies \quad \mu - 1.2816\sigma = 780. \tag{1}$$

Similarly,

$$P(X > 830) = 0.20 \implies P(X \le 830) = 0.80 \implies \Phi\left(\frac{830 - \mu}{\sigma}\right) = 0.80.$$

From the table, $\Phi(0.8416) \approx 0.80$, so

$$\frac{830 - \mu}{\sigma} = 0.8416 \quad \implies \quad \mu + 0.8416\sigma = 830. \tag{2}$$

Subtracting (1) from (2):

$$(0.8416 + 1.2816)\sigma = 50 \implies 2.1232\,\sigma = 50 \implies \sigma \approx 23.55.$$

Substituting back into (2): $\mu = 830 - 0.8416(23.55) \approx 810.2$.
Hence the mean mass is about $\mu \approx 810$ g and the standard deviation $\sigma \approx 23.6$ g.

4.8 Linear Interpolation Between Table Entries (Useful Extra)

The table gives z only to two decimal places. If a calculation produces, say, $z = 1.645$, the honest table reading is ambiguous between $\Phi(1.64) = 0.9495$ and $\Phi(1.65) = 0.9505$. **Linear interpolation** estimates the value in between by assuming the curve is approximately straight over the small interval:

$$\Phi(1.645) \approx 0.9495 + \frac{1.645 - 1.64}{1.65 - 1.64} \times (0.9505 - 0.9495) = 0.9495 + 0.5 \times 0.0010 = 0.9500.$$

In the GCE, examiners usually accept the nearest table entry, but interpolation gives extra accuracy and shows real mastery. The same idea works in reverse: to find z from $\Phi(z) = p$ when p falls strictly between two entries, interpolate between the corresponding z-values.

KEY POINTS **Key points of this section**

- $Z \sim N(0,1)$ is the standard normal variable; $\Phi(z) = P(Z \le z)$ is tabulated for $z \ge 0$.
- Standardisation: if $X \sim N(\mu, \sigma^2)$ then $Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$, and $x = \mu + \sigma z$.
- Core formulas: $P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$; $P(X \ge x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$; $P(a \le X \le b) = \Phi(z_b) - \Phi(z_a)$.
- Symmetry: $\Phi(-z) = 1 - \Phi(z)$ — never $-\Phi(z)$.
- Inverse problems: read the table backwards to get z from p, then $x = \mu + \sigma z$; two probability statements give two simultaneous equations for unknown μ and σ.

EXERCISE 5 **Practice**

1. Given $Z \sim N(0,1)$, find (a) $P(Z \le 2.35)$; (b) $P(Z > -1.04)$; (c) $P(0.5 \le Z \le 1.87)$; (d) $P(|Z| > 1.5)$.
2. The heights of students in a school in Bamenda are modelled by $H \sim N(168, 6^2)$ cm. Find the probability that a randomly chosen student is (a) taller than 180 cm; (b) between 160 cm and 175 cm.
3. For $X \sim N(70, 25)$, find the value x such that (a) $P(X \le x) = 0.90$; (b) $P(X \ge x) = 0.025$.
4. The lifetime T (in hours) of an electric bulb is normal. Given $P(T < 950) = 0.1587$ and $P(T > 1080) = 0.0228$, find the mean and standard deviation of T.

SOLUTION **Exercise 1**

1. (a) $\Phi(2.35) = 0.9906$. (b) $P(Z > -1.04) = \Phi(1.04) = 0.8508$. (c) $\Phi(1.87) - \Phi(0.5) = 0.9693 - 0.6915 = 0.2778$. (d) $P(|Z| > 1.5) = 2\big(1 - \Phi(1.5)\big) = 2(1 - 0.9332) = 0.1336$.
2. (a) $z = \frac{180-168}{6} = 2$; $P(H > 180) = 1 - \Phi(2) = 0.0228$. (b) $z_1 = \frac{160-168}{6} = -1.33$, $z_2 = \frac{175-168}{6} = 1.17$; probability $= \Phi(1.17) - \Phi(-1.33) = 0.8790 - (1 - 0.9082) = 0.7872$.
3. (a) $\Phi(z) = 0.90 \Rightarrow z = 1.2816$; $x = 70 + 5(1.2816) = 76.4$. (b) $P(X \le x) = 0.975 \Rightarrow z = 1.96$; $x = 70 + 5(1.96) = 79.8$.
4. $\Phi(-1) = 0.1587$ and $\Phi(2) = 0.9772$, so $\frac{950-\mu}{\sigma} = -1$ and $\frac{1080-\mu}{\sigma} = 2$. Subtracting: $3\sigma = 130$, so $\sigma \approx 43.3$ hours and $\mu = 950 + 43.3 \approx 993.3$ hours.

DID YOU KNOW? **Why 1.96?**

The value $z = 1.96$ is famous worldwide: since $\Phi(1.96) = 0.975$, exactly 95% of any normal population lies within 1.96 standard deviations of the mean. Doctors, engineers and pollsters use this every day when they build "95% confidence intervals". The number you just computed in this section is the backbone of modern data analysis — including the quality control of goods produced in Cameroonian factories.

5 NORMAL APPROXIMATIONS TO THE BINOMIAL AND POISSON DISTRIBUTIONS

In the previous section we mastered the standard normal variable $Z \sim N(0,1)$ and learned to read probabilities from the standard normal table. We now put that tool to work: when a discrete distribution (binomial or Poisson) involves a large parameter, exact calculations become prohibitively heavy. The **Central Limit Theorem** guarantees that the shape of the distribution approaches a bell curve, so we can replace the discrete histogram by a continuous normal density — provided we apply a **continuity correction** to account for the jump from integers to real numbers.

PROPERTY **Central Limit Theorem (Informal Statement)**

If $X_1, X_2, \ldots, X_n$ are independent and identically distributed random variables with mean μ and variance σ^2, then the sum $S_n = \sum_{i=1}^{n} X_i$ (and the sample mean $\bar{X}$) is approximately normally distributed for large n:

$$S_n \dot{\sim} N(n\mu,\, n\sigma^2), \qquad \bar{X} \dot{\sim} N\left(\mu,\, \frac{\sigma^2}{n}\right).$$

Both the Binomial $B(n,p)$ (sum of n Bernoulli trials) and the Poisson $Po(\lambda)$ (limit of Binomial, or sum of many rare events) fall under this theorem.

5.1 Why Approximate?

Consider $X \sim B(100, 0.55)$. To find $P(X \geq 60)$ exactly we would need to sum 41 binomial terms:

$$P(X \geq 60) = \sum_{k=60}^{100} \binom{100}{k} (0.55)^k (0.45)^{100-k}.$$

Even with a calculator this is tedious. Similarly, for $X \sim Po(25)$, $P(X \leq 20) = \sum_{k=0}^{20} e^{-25}\frac{25^k}{k!}$ requires 21 terms. The normal distribution offers a fast, accurate alternative when the parameters are large enough. The approximations we study here are direct consequences of the Central Limit Theorem and are standard tools in the GCE Advanced Level examination.

5.2 Normal Approximation to the Binomial Distribution

PROPERTY **Normal Approximation to the Binomial**

Let $X \sim B(n, p)$ with n large and p not too close to 0 or 1. A common rule of thumb (examinable) is:

$$np > 5 \quad \text{and} \quad n(1-p) > 5.$$

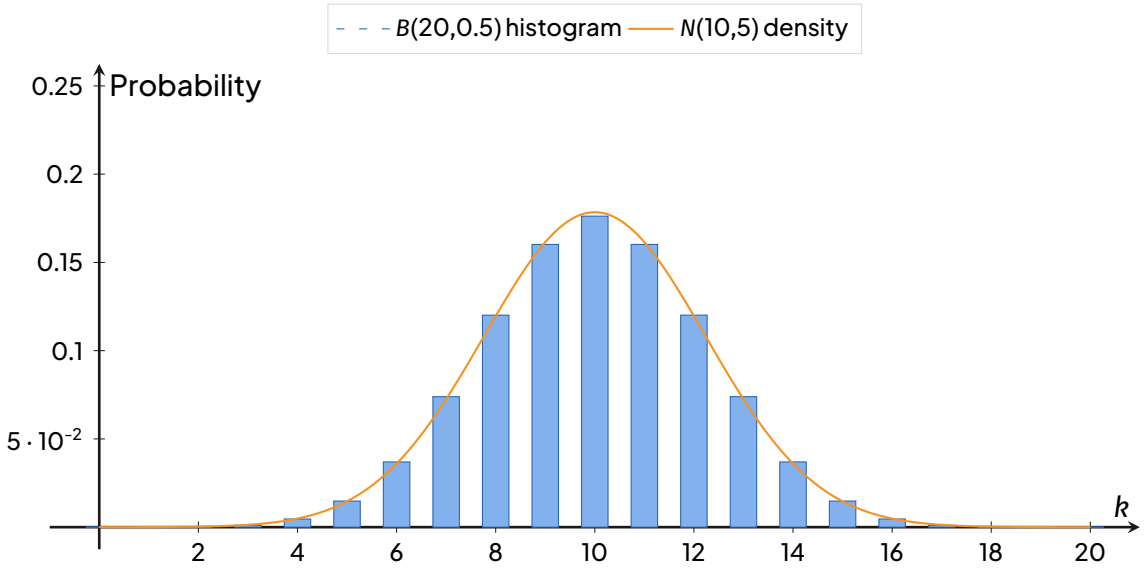
Then X can be approximated by a normal distribution:

$$X \dot{\sim} N\left(\mu = np,\ \sigma^2 = np(1-p)\right).$$

The symbol $\dot{\sim}$ means "is approximately distributed as".

WARNING **Validity Conditions**

Always state and check the conditions $np > 5$ and $n(1-p) > 5$ before using the approximation. Failure to do so loses marks in the GCE.



Histogram of $B(20, 0.5)$ with superimposed normal curve $N(10, 5)$. The fit is already good at $n = 20$.

5.3 Continuity Correction for the Binomial

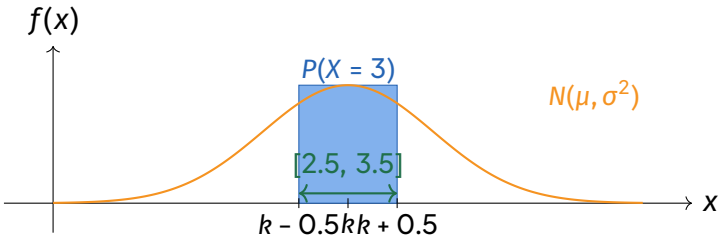
Because the binomial variable takes only integer values while the normal variable is continuous, we must replace each integer k by the interval $[k - 0.5,\ k + 0.5]$. The area under the normal curve over that interval approximates the height of the histogram bar at k.

METHOD **Continuity Correction — Binomial**

To approximate $P(X \text{ relation } k)$ where $X \sim B(n, p)$ and k is an integer:

1. Write the event in terms of $\leq$ or $\geq$ (or $=$).
2. Replace the discrete bound by a continuous interval using ± 0.5:
   - $P(X \leq k) \approx P(Y \leq k + 0.5)$
   - $P(X < k) = P(X \leq k - 1) \approx P(Y \leq k - 0.5)$
   - $P(X \geq k) \approx P(Y \geq k - 0.5)$
   - $P(X > k) = P(X \geq k + 1) \approx P(Y \geq k + 0.5)$
   - $P(X = k) \approx P(k - 0.5 \leq Y \leq k + 0.5)$

   where $Y \sim N(np,\ np(1-p))$.
3. Standardise: $Z = \dfrac{Y - \mu}{\sigma}$ and use the standard normal table.



Continuity correction: the discrete probability $P(X = k)$ (blue bar) is approximated by the area under the normal curve (orange) from $k - 0.5$ to $k + 0.5$ (green interval).

WORKED EXAMPLE 12 **Germination of Maize Seeds**

An agricultural station in Buea tests a new variety of maize with a germination probability of 0.55. A farmer plants 100 seeds. Use a normal approximation to find the probability that at least 60 seeds germinate.

SOLUTION

Solution. Let X be the number of seeds that germinate. $X \sim B(100, 0.55)$. 1. Check conditions: $np = 100 \times 0.55 = 55 > 5$, $n(1-p) = 100 \times 0.45 = 45 > 5$. ✓2. Normal approximation: $X \therefore Y \sim N(\mu = 55,\ \sigma^2 = 100 \times 0.55 \times 0.45 = 24.75)$, $\sigma = \sqrt{24.75} \approx 4.975$. 3. Continuity correction: "At least 60" means $X \geq 60$. We use $P(X \geq 60) \approx P(Y \geq 59.5)$. 4. Standardise:

$$z = \frac{59.5 - 55}{\sqrt{24.75}} = \frac{4.5}{4.975} \approx 0.9045.$$

5. Use standard normal table: $P(Y \geq 59.5) = P(Z \geq 0.9045) = 1 - \Phi(0.9045)$. From tables, $\Phi(0.90) = 0.8159$, $\Phi(0.91) = 0.8186$. Interpolating: $\Phi(0.9045) \approx 0.8171$.

$$P(X \geq 60) \approx 1 - 0.8171 = 0.1829.$$

Answer: The approximate probability is **0.183** (3 s.f.).

5.4 Normal Approximation to the Poisson Distribution

PROPERTY **Normal Approximation to the Poisson**

Let $X \sim Po(\lambda)$ with λ large. A common rule of thumb (examinable) is:

$$\lambda > 10.$$

Then X can be approximated by a normal distribution:

$$X \dot{\sim} N(\mu = \lambda,\ \sigma^2 = \lambda).$$

WARNING **Validity Condition**

Always state $\lambda > 10$ before approximating a Poisson by a normal. For $\lambda \leq 10$ the approximation is poor; use exact Poisson tables or the binomial approximation (if applicable).

The continuity correction works exactly the same way because the Poisson variable is also discrete (integer-valued).

METHOD **Continuity Correction — Poisson**

For $X \sim Po(\lambda)$ approximated by $Y \sim N(\lambda, \lambda)$:

- $P(X \leq k) \approx P(Y \leq k + 0.5)$
- $P(X < k) \approx P(Y \leq k - 0.5)$
- $P(X \geq k) \approx P(Y \geq k - 0.5)$
- $P(X > k) \approx P(Y \geq k + 0.5)$
- $P(X = k) \approx P(k - 0.5 \leq Y \leq k + 0.5)$

Then standardise $Z = \dfrac{Y - \lambda}{\sqrt{\lambda}}$ and read from the normal table.

WORKED EXAMPLE 13 **Call Centre — CAMTEL Douala**

A CAMTEL call centre in Douala receives an average of 25 calls per hour. Assuming a Poisson model, use a normal approximation to find the probability that in a given hour the centre receives at most 20 calls.

SOLUTION

Solution. Let X be the number of calls in one hour. $X \sim Po(25)$. 1. Check condition: $\lambda = 25 > 10$. ✓ 2. Normal approximation: $X \overset{\cdot}{\sim} Y \sim N(\mu = 25,\ \sigma^2 = 25)$, so $\sigma = 5$. 3. Continuity correction: "At most 20" means $X \le 20$. We use $P(X \le 20) \approx P(Y \le 20.5)$. 4. Standardise:

$$z = \frac{20.5 - 25}{5} = \frac{-4.5}{5} = -0.9.$$

5. Use standard normal table: $P(Y \le 20.5) = P(Z \le -0.9) = \Phi(-0.9) = 1 - \Phi(0.9)$. From tables, $\Phi(0.9) = 0.8159$.

$$P(X \le 20) \approx 1 - 0.8159 = 0.1841.$$

Answer: The approximate probability is **0.184** (3 s.f.).

5.5 Choosing the Correct Continuity Correction

This is a classic GCE trap. The key is to rewrite the required probability in terms of ≤ or ≥ before applying the ±0.5 shift.

KEY POINTS **Continuity Correction Quick Reference**

Required probability	Rewrite as	Normal interval
$P(X \le k)$	$P(X \le k)$	$P(Y \le k + 0.5)$
$P(X < k)$	$P(X \le k - 1)$	$P(Y \le k - 0.5)$
$P(X \ge k)$	$P(X \ge k)$	$P(Y \ge k - 0.5)$
$P(X > k)$	$P(X \ge k + 1)$	$P(Y \ge k + 0.5)$
$P(X = k)$	$P(X = k)$	$P(k - 0.5 \le Y \le k + 0.5)$

WARNING **Common Mistakes**

- Forgetting the continuity correction entirely (the #1 cause of lost marks).
- Applying the correction in the wrong direction (e.g. using $k + 0.5$ for $X \geq k$).
- Not rewriting strict inequalities ($<$, $>$) in terms of $\leq$ or $\geq$ first.
- Using the normal approximation without checking the validity conditions ($np > 5$, $n(1-p) > 5$, $\lambda > 10$).

5.6 Summary of Approximations (Synthesis)

You have now seen three approximations across the Further Mathematics course. The table below summarises them for revision.

Approximation	Original Distribution	Approximating Distribution	Validity Conditions
Poisson → Binomial	$B(n, p)$ with n large, p small	$Po(\lambda = np)$	$n \geq 50$, $p \leq 0.1$, $np \leq 5$ (Lower Sixth)
Normal → Binomial	$B(n, p)$	$N(np,\ np(1-p))$	$np > 5$ and $n(1-p) > 5$
Normal → Poisson	$Po(\lambda)$	$N(\lambda,\ \lambda)$	$\lambda > 10$

DID YOU KNOW? **Historical Note**

The normal approximation to the binomial was first derived by Abraham de Moivre in 1733 (published in The Doctrine of Chances). He used it to avoid summing huge binomial coefficients — exactly the problem we face today! The continuity correction was later refined by Laplace and Gauss.

KEY POINTS **Exam Tip — The Five-Line Chain**

In the GCE, present your solution as five clear lines:

1. Distribution: $X \sim B(n, p)$ or $X \sim Po(\lambda)$.
2. Check conditions: $np > 5$, $n(1-p) > 5$ or $\lambda > 10$.
3. Approximation: $X \dot{\sim} Y \sim N(\mu, \sigma^2)$.
4. Continuity correction: Rewrite the probability and apply ± 0.5.
5. Standardise & table: $Z = \frac{Y-\mu}{\sigma}$, read $\Phi(z)$.

This structured approach guarantees you earn all method marks even if the final arith-

metic has a slip.

EXERCISE 6 **Practice**

1. A factory in Douala produces light bulbs with a defect rate of 2%. In a batch of 500 bulbs, use a normal approximation to find the probability that there are more than 15 defective bulbs.
2. The number of accidents per month at a certain junction in Yaoundé follows a Poisson distribution with mean 12. Use a normal approximation to find the probability that in a given month there are between 8 and 15 accidents inclusive.
3. A multiple-choice test has 80 questions, each with 4 options. A student guesses all answers. Use a normal approximation to find the probability that the student gets at least 25 correct answers.

SOLUTION **Exercise 1**

$X \sim B(500, 0.02)$. $np = 10 > 5$, $n(1-p) = 490 > 5$. $X \dot{\sim} Y \sim N(10, 9.8)$, $\sigma \approx 3.1305$. $P(X > 15) = P(X \geq 16) \approx P(Y \geq 15.5)$. $z = \frac{15.5-10}{3.1305} \approx 1.757$. $P(Z \geq 1.757) = 1 - \Phi(1.757) \approx 1 - 0.9605 = 0.0395$.

SOLUTION **Exercise 2**

$X \sim Po(12)$. $\lambda = 12 > 10$. $X \dot{\sim} Y \sim N(12, 12)$, $\sigma = \sqrt{12} \approx 3.464$. $P(8 \leq X \leq 15) \approx P(7.5 \leq Y \leq 15.5)$. $z_1 = \frac{7.5-12}{3.464} \approx -1.30$, $z_2 = \frac{15.5-12}{3.464} \approx 1.01$. $P(-1.30 \leq Z \leq 1.01) = \Phi(1.01) - \Phi(-1.30) = 0.8438 - (1 - 0.9032) = 0.7470$.

SOLUTION **Exercise 3**

$X \sim B(80, 0.25)$. $np = 20 > 5$, $n(1-p) = 60 > 5$. $X \dot{\sim} Y \sim N(20, 15)$, $\sigma = \sqrt{15} \approx 3.873$. $P(X \geq 25) \approx P(Y \geq 24.5)$. $z = \frac{24.5-20}{3.873} \approx 1.162$. $P(Z \geq 1.162) = 1 - \Phi(1.162) \approx 1 - 0.8770 = 0.1230$.

6 INTEGRATION ACTIVITY

7 INTEGRATION ACTIVITY: SPECIAL CONTINUOUS DISTRIBUTIONS

7.1 Aims of the activity

This integration activity closes the chapter by bringing together, in a single realistic study, all the special continuous distributions you have met: the **exponential distribution**, the **uniform (rectangular) distribution**, the **normal distribution**, and the **normal approximation** to a discrete distribution with **continuity correction**.

By the end of the activity, you should be able to:

- choose, and justify the choice of, an appropriate probability model for a real situation;
- use $P(X > x) = e^{-\lambda x}$ for an exponential model and interpret the parameter λ;
- compute the mean $\frac{a+b}{2}$ and variance $\frac{(b-a)^2}{12}$ of a uniform distribution on $[a, b]$;
- standardise a normal variable using $Z = \frac{X-\mu}{\sigma}$ and read probabilities from the standard normal table, including inverse readings (finding a value from a probability);
- check the validity conditions ($np > 5$, $nq > 5$; $\lambda > 10$) before replacing a binomial or Poisson distribution by a normal one, and apply the continuity correction correctly;
- communicate a complete statistical argument in a group presentation, as expected in the GCE Advanced Level examination.

METHOD **How to work**

Form groups of three or four. Each group keeps a written record of: the model chosen, the reason for the choice, the calculations, and the conclusion stated in everyday language. Allow about 15 minutes per task, then 15 minutes for the group presentations. The standard normal table and a non-programmable calculator are allowed.

7.2 Situation

The **SONEL** water-pumping station of Bonabéri, in Douala, supplies a residential neighbourhood. The station manager, Mrs Etame, collects data in order to plan maintenance and billing. Her records give the following information.

Breakdowns. Breakdowns of the main pump occur at random, independently of one another, at a constant average rate of **one breakdown every 40 days**. The time T (in days) between two

consecutive breakdowns may therefore be modelled by an exponential distribution.

Repair crew. When a breakdown is reported by telephone, the repair crew's arrival time U (in hours) after the call is equally likely to be anywhere between 1 hour and 3 hours. It may therefore be modelled by a uniform distribution on $[1, 3]$.

Consumption. The daily water consumption X of the neighbourhood (in m^3) is known from long experience to be normally distributed with mean $\mu = 850\ \mathrm{m}^3$ and standard deviation $\sigma = 60\ \mathrm{m}^3$.

Bill payment. The neighbourhood contains 200 households supplied by the station. Each household, independently of the others, pays its monthly bill on time with probability 0.6. Let Y be the number of households paying on time in a given month.

Mrs Etame asks your group of Upper Sixth Further Mathematics students to answer four questions that she needs for her annual report:

1. What is the probability that the pump runs for more than 60 days without any breakdown?
2. What are the mean and the variance of the crew's arrival time?
3. What is the probability that the daily consumption exceeds $950\ \mathrm{m}^3$? Above what level is the consumption on only 5% of days (the station must be able to supply this level)?
4. What is the probability that at least 110 households pay their bill on time in a given month?

7.3 Tasks

Task 1 — Modelling the breakdowns (exponential distribution).

1. Explain why the exponential distribution is a reasonable model for T. Which property of the breakdowns justifies this choice?
2. Show that the rate parameter is $\lambda = \dfrac{1}{40} = 0.025$ per day, and write down the probability density function of T.
3. Compute $P(T > 60)$. Give your answer to 3 decimal places and interpret it in one sentence for Mrs Etame.
4. Compute the mean time between breakdowns and check that it agrees with the data.

Task 2 — Modelling the repair crew (uniform distribution).

1. Write down the probability density function of U on $[1, 3]$ and sketch its graph.
2. Compute $E(U)$ and $\mathrm{Var}(U)$ using the formulae for the uniform distribution.
3. Compute $P(U < 1.5)$, the probability that the crew arrives within an hour and a half.

Task 3 — Modelling the consumption (normal distribution).

1. Standardise X and compute $P(X > 950)$, giving your answer to 4 decimal places.
2. Let c be the consumption level exceeded on only 5% of days. Write the equation $P(X > c) = 0.05$ in terms of the standard normal variable Z, find the corresponding value z from the table (use $z \approx 1.645$), and deduce c to the nearest m^3.

3. Explain in one sentence why this value c matters for the sizing of the pump.

Task 4 — Modelling bill payments (normal approximation to the binomial).

1. State the exact distribution of Y and its parameters.
2. Verify that the conditions for a normal approximation are satisfied, and give the mean and variance of the approximating normal distribution.
3. Using the continuity correction, compute an approximate value of $P(Y \geq 110)$.
4. Justify why the continuity correction is needed here, and say what would go wrong without it.

Task 5 — Group presentation.

Prepare a five-minute presentation of your four models. For each one, state: the distribution chosen, the reason for the choice, the validity conditions checked, the numerical result, and a conclusion in plain language that Mrs Etame — who is not a mathematician — can understand.

7.4 Model answer

Task 1. Breakdowns occur at random, independently, at a constant average rate: these are exactly the conditions under which the waiting time between events follows an exponential distribution (the continuous analogue of the geometric distribution, with the memoryless property).

The mean of an exponential distribution is $\frac{1}{\lambda}$, and the mean time between breakdowns is 40 days, so

$$\frac{1}{\lambda} = 40 \quad \Longrightarrow \quad \lambda = \frac{1}{40} = 0.025 \text{ per day.}$$

The probability density function is

$$f(t) = 0.025\, e^{-0.025t}, \quad t \geq 0 \quad \text{(and 0 otherwise).}$$

Then

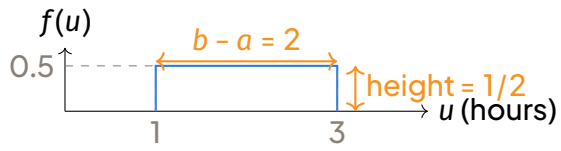
$$P(T > 60) = e^{-\lambda \times 60} = e^{-0.025 \times 60} = e^{-1.5} \approx 0.223.$$

Interpretation: there is about a 22% chance that the pump runs for more than 60 days without any breakdown. The mean is $E(T) = \frac{1}{\lambda} = 40$ days, which matches the data, as expected.

Task 2. The arrival time is uniform on $[1, 3]$, so

$$f(u) = \frac{1}{3 - 1} = \frac{1}{2}, \quad 1 \leq u \leq 3,$$

and $f(u) = 0$ otherwise. The graph is a horizontal rectangle of height 0.5 between $u = 1$ and $u = 3$ (total area = 1).



Probability density function of U ~ Uniform[1, 3]

Mean and variance:

$$E(U) = \frac{a+b}{2} = \frac{1+3}{2} = 2 \text{ hours}, \qquad \operatorname{Var}(U) = \frac{(b-a)^2}{12} = \frac{(3-1)^2}{12} = \frac{4}{12} = \frac{1}{3} \text{ hours}^2.$$

The standard deviation is $\sqrt{1/3} \approx 0.577$ hour, about 35 minutes.

Probability of arrival within 1.5 hours:

$$P(U < 1.5) = \frac{1.5-1}{3-1} = \frac{0.5}{2} = 0.25.$$

Task 3. Here $X \sim N(850,\ 60^2)$. Standardising with $Z = \dfrac{X-850}{60}$:

$$P(X > 950) = P\left(Z > \frac{950-850}{60}\right) = P(Z > 1.6667).$$

From the standard normal table, $\Phi(1.67) \approx 0.9525$, so

$$P(X > 950) = 1 - 0.9525 = 0.0475.$$

So consumption exceeds 950 m^3 on roughly 4.75% of days, about one day in twenty.

For the level c exceeded on only 5% of days:

$$P(X > c) = 0.05 \iff P\left(Z > \frac{c-850}{60}\right) = 0.05 \iff \Phi\left(\frac{c-850}{60}\right) = 0.95.$$

The table gives $\Phi(1.645) \approx 0.95$, hence

$$\frac{c-850}{60} = 1.645 \quad \Longrightarrow \quad c = 850 + 1.645 \times 60 = 948.7 \approx 949 \text{ m}^3.$$

Interpretation: the pump and pipes must be able to deliver at least about 949 m^3 per day if the station is to meet demand on 95% of days.

Task 4. Exactly, $Y \sim B(200,\ 0.6)$, since there is a fixed number of independent households, each with the same probability $p = 0.6$ of paying on time.

Validity conditions:

$$np = 200 \times 0.6 = 120 > 5, \qquad nq = 200 \times 0.4 = 80 > 5,$$

so the normal approximation is appropriate. The approximating distribution is

$$Y \approx N(np,\ npq) = N(120,\ 48), \qquad \sigma = \sqrt{48} \approx 6.928.$$

"At least 110" means $Y \geq 110$, i.e. $Y = 110, 111, \ldots, 200$. With the continuity correction, the integer 110 is represented by the interval [109.5, 110.5], so

$$P(Y \geq 110) \approx P(Y_N \geq 109.5) = P\left(Z \geq \frac{109.5 - 120}{6.928}\right) = P(Z \geq -1.516).$$

By symmetry and the table,

$$P(Z \geq -1.516) = \Phi(1.516) \approx 0.935.$$

So there is approximately a 93.5% chance that at least 110 households pay on time.

The continuity correction is essential because we approximate a discrete variable (whole households) by a continuous one: without it we would compute $P(Z \geq -1.66)$ using the boundary 110 itself, which treats the integer 110 as a point of zero width and systematically distorts the probability.

Task 5 (guidance). A good presentation states, for each model: exponential because breakdowns are random, independent events at a constant rate; uniform because every arrival time in [1, 3] is equally likely; normal because consumption results from many small independent influences; normal approximation because n is large with np and nq both greater than 5, with continuity correction because the binomial is discrete.

WARNING **Pitfalls to avoid in this activity**

- Do not confuse λ with the mean: for the exponential distribution the mean is $\frac{1}{\lambda}$, here 40, not $\lambda = 40$.
- $P(T > 60) = e^{-\lambda \times 60}$: remember that λ must be in matching units (per day, since 60 is in days).
- For the uniform distribution, $\mathrm{Var}(U) = \frac{(b-a)^2}{12}$, not $\frac{(b-a)^2}{2}$.
- In Task 3(b), the 5% is in the upper tail, so use $\Phi(z) = 0.95$, i.e. $z = +1.645$, not -1.645.
- Never apply the normal approximation without first checking $np > 5$ and $nq > 5$; and never forget the continuity correction when moving from a discrete to a continuous model.

DID YOU KNOW? **Why this matters in Cameroon**

Water and electricity utilities such as ENEO and Camwater plan maintenance teams, reservoir sizes and billing forecasts using exactly these tools: exponential models for time between failures, uniform models when only bounds are known, and normal models for aggregated consumption. Mastering them opens the way to careers in engineering, actuarial science and data analysis — sectors that are growing rapidly in Douala and Yaoundé.

8 METHODS AND WORKED EXAMPLES

9 SPECIAL CONTINUOUS DISTRIBUTIONS

9.1 Introduction and Overview

In the previous chapter we studied the general theory of continuous random variables: probability density functions, cumulative distribution functions, expectation and variance. We now turn to three specific distributions that appear constantly in modelling real phenomena — the exponential, the uniform (rectangular) and the normal distributions. The first two have simple algebraic forms that allow exact calculations; the normal distribution, while not integrable in elementary terms, is the cornerstone of statistical inference because of the Central Limit Theorem. We shall also learn how the normal distribution can approximate the binomial and Poisson distributions when the parameters are large, a technique that remains essential for GCE examination questions.

DID YOU KNOW? **Why these three?**

The exponential distribution models waiting times between random events (phone calls arriving at a switchboard, radioactive decays, component lifetimes). The uniform distribution models complete ignorance over a finite interval (a random arrival time, a rounding error). The normal distribution arises whenever many small independent effects add up (heights, measurement errors, examination marks). Together they cover a vast range of applications in engineering, economics, biology and the social sciences.

9.2 The Exponential Distribution

DEFINITION **Exponential distribution**

A continuous random variable X follows an exponential distribution with rate parameter $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$, if its probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

The cumulative distribution function is

$$F(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

PROPERTY **Mean and variance of the exponential distribution**

For $X \sim \text{Exp}(\lambda)$:

$$E(X) = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}, \qquad \sigma = \frac{1}{\lambda}.$$

Proof (examinable).

$$E(X) = \int_0^\infty x\lambda e^{-\lambda x}\,dx = \lambda\left[-\frac{x}{\lambda}e^{-\lambda x}\right]_0^\infty + \int_0^\infty \frac{1}{\lambda}e^{-\lambda x}\,dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

$$E(X^2) = \int_0^\infty x^2\lambda e^{-\lambda x}\,dx = \frac{2}{\lambda^2} \quad \text{(integration by parts twice).}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

PROPERTY **Memoryless property**

The exponential distribution is the only continuous distribution with the memoryless property: for all $s, t \geq 0$,

$$P(X > s + t \mid X > s) = P(X > t) = e^{-\lambda t}.$$

This means that a component that has already survived s hours is as good as new — its remaining lifetime has exactly the same exponential distribution.

METHOD **Handling an exponential model**

1. Identify the parameter λ. If the mean μ is given, $\lambda = 1/\mu$. If λ is given directly, $\mu = 1/\lambda$ and $\sigma^2 = 1/\lambda^2$.
2. Use the survival function $P(X > x) = e^{-\lambda x}$ for $x \geq 0$. This is faster than integrating the pdf.
3. For an interval $a < X < b$ ($0 \leq a < b$): $P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$.
4. Apply the memoryless property for conditional probabilities of the form $P(X > s + t \mid X > s)$.
5. Check units: x and λ must use the same time unit (both in minutes, or both in hours, etc.).

EXEMPLE **Lifetime of an electronic component**

The lifetime T (in hours) of an electronic component used in a phone repair shop in Douala

is modelled by an exponential distribution with mean 500 hours.

(a) Find the probability that a component lasts more than 600 hours.

(b) Find the probability that a component lasts between 200 and 400 hours.

(c) Given that a component has already worked for 300 hours, find the probability that it will work for at least 200 more hours.

(d) Find the median lifetime.

WORKED EXAMPLE 14 **Lifetime of an electronic component**

Solution. Mean = 500 hours $\Rightarrow \lambda = \dfrac{1}{500} = 0.002$ per hour.

(a) $P(T > 600) = e^{-\lambda \times 600} = e^{-0.002 \times 600} = e^{-1.2} \approx 0.3012$.

(b) $P(200 < T < 400) = e^{-0.002\times200} - e^{-0.002\times400} = e^{-0.4} - e^{-0.8}$. Numerically, $e^{-0.4} \approx 0.6703$, $e^{-0.8} \approx 0.4493$, so $P \approx 0.6703 - 0.4493 = 0.2210$.

(c) By the memoryless property: $P(T > 300 + 200 \mid T > 300) = P(T > 200) = e^{-0.002\times200} = e^{-0.4} \approx 0.6703$.

(d) Median m satisfies $P(T \le m) = \frac{1}{2}$: $1 - e^{-0.002m} = \frac{1}{2} \implies e^{-0.002m} = \frac{1}{2} \implies 0.002\,m = \ln 2$. Hence $m = \dfrac{\ln 2}{0.002} = 500 \ln 2 \approx 346.6$ hours. Note: median (346.6 h) < mean (500 h) — typical for a right-skewed distribution.

WARNING **Common mistakes with the exponential distribution**

- Confusing λ (rate) with $1/\lambda$ (mean). Always write $\lambda = 1/\text{mean}$ explicitly.
- Forgetting that the pdf is zero for $x < 0$; probabilities for negative values are zero.
- Using the memoryless property incorrectly: it applies only to $P(X > s + t \mid X > s)$, not to $P(X < s + t \mid X < s)$.
- Mixing units (e.g., λ in per hour but x in minutes). Convert everything to the same unit first.

9.3 The Uniform (Rectangular) Distribution

DEFINITION **Uniform distribution on $[a, b]$**

A continuous random variable X follows a uniform (or rectangular) distribution on the in-

terval $[a, b]$ $(a < b)$, written $X \sim U(a, b)$, if its probability density function is

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The cumulative distribution function is

$$F(x) = P(X \le x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

PROPERTY **Mean and variance of the uniform distribution**

For $X \sim U(a, b)$:

$$E(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}, \qquad \sigma = \frac{b-a}{\sqrt{12}}.$$

Proof (examinable).

$$E(X) = \int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}.$$

$$E(X^2) = \int_a^b x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}.$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}.$$

METHOD **Handling a uniform (rectangular) model**

1. Write down the key results: $E(X) = \dfrac{a+b}{2}$, $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$, $\sigma = \dfrac{b-a}{\sqrt{12}}$.
2. Probabilities are ratios of lengths (areas of rectangles): $P(c < X < d) = \dfrac{d-c}{b-a}$ for $a \le c < d \le b$.
3. If an endpoint is unknown, set up an equation from the given mean or variance and solve for a or b.
4. Always sketch the rectangle of height $\frac{1}{b-a}$: the required probability is the shaded area.

EXEMPLE **Waiting time at a bus stop**

Buses leave a motor park in Bamenda every 20 minutes. A passenger arrives at random. Her waiting time X (in minutes) is uniformly distributed on $[0, 20]$.

(a) Write down the probability density function of X.

(b) Calculate the mean and the standard deviation of the waiting time.

(c) Find the probability that she waits more than 12 minutes.

(d) Find the probability that she waits between 5 and 15 minutes.

WORKED EXAMPLE 15 **Waiting time at a bus stop**

Solution. Here $a = 0$, $b = 20$, so $b - a = 20$.

(a) $f(x) = \begin{cases} \frac{1}{20} & 0 \le x \le 20, \\ 0 & \text{otherwise.} \end{cases}$

(b) $E(X) = \frac{0+20}{2} = 10$ minutes. $\text{Var}(X) = \frac{(20-0)^2}{12} = \frac{400}{12} = \frac{100}{3}$. $\sigma = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} \approx 5.77$ minutes.

(c) $P(X > 12) = \frac{20-12}{20} = \frac{8}{20} = 0.4$.

(d) $P(5 < X < 15) = \frac{15-5}{20} = \frac{10}{20} = 0.5$. Each answer is the area of a rectangle of height $\frac{1}{20}$ over the relevant interval.

EXEMPLE **Finding an unknown endpoint**

A continuous random variable X is uniformly distributed on $[2, b]$ with $b > 2$. Given that $\text{Var}(X) = 12$, find

(a) the value of b;

(b) $E(X)$ and $P(X > 5)$.

WORKED EXAMPLE 16 **Finding an unknown endpoint**

Solution. (a) For $X \sim U(2, b)$: $\text{Var}(X) = \frac{(b-2)^2}{12} = 12 \implies (b-2)^2 = 144 \implies b - 2 = 12$ (positive root since $b > 2$). Hence $\boxed{b = 14}$.

(b) $E(X) = \frac{2+14}{2} = 8$. Density height $= \frac{1}{14-2} = \frac{1}{12}$. $P(X > 5) = \frac{14-5}{12} = \frac{9}{12} = 0.75$.

WARNING **Common mistakes with the uniform distribution**

- Using $\text{Var}(X) = \frac{(b-a)^2}{2}$ instead of $\frac{(b-a)^2}{12}$.
- Forgetting that the pdf is constant only on $[a, b]$; outside this interval it is zero.
- In probability calculations, using the wrong interval length (e.g., using $b - a$ instead of $d - c$).
- Not checking that the computed a and b satisfy $a < b$.

9.4 The Normal Distribution

DEFINITION **Normal distribution**

A continuous random variable X follows a normal distribution with mean μ and variance σ^2 ($\sigma > 0$), written $X \sim N(\mu, \sigma^2)$, if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \qquad -\infty < x < \infty.$$

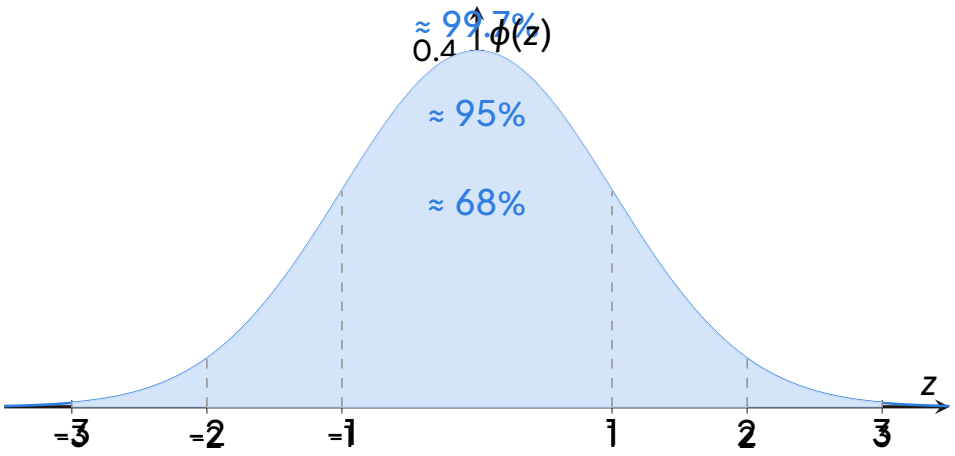
The curve is symmetric about $x = \mu$, bell-shaped, with inflection points at $x = \mu \pm \sigma$.

PROPERTY **Mean and variance**

For $X \sim N(\mu, \sigma^2)$: $E(X) = \mu$, $\text{Var}(X) = \sigma^2$. Proof (not examinable in full, but the symmetry argument for the mean is expected). By symmetry of $f(x)$ about μ, $E(X) = \mu$. The variance derivation uses the substitution $z = \frac{x-\mu}{\sigma}$ and the standard Gaussian integral $\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\,dz = \sqrt{2\pi}$.

KEY POINTS **Key features of the normal curve**

- Total area under the curve = 1.
- Symmetric about μ: $P(X < \mu) = P(X > \mu) = 0.5$.
- Approximately 68% of the area lies within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$ (empirical rule).
- The standard normal distribution is $Z \sim N(0, 1)$ with pdf $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and cdf $\Phi(z) = P(Z \le z)$.



Standard normal density $\phi(z)$ with empirical rule areas

9.5 The Standard Normal Variable and Use of Tables

PROPERTY **Standardisation**

If $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X - \mu}{\sigma} \sim N(0, 1)$. For any x,

$$P(X < x) = P\left(Z < \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right),$$

where $\Phi(z)$ is the cumulative distribution function of the standard normal variable, tabulated for $z \geq 0$.

METHOD **Computing $P(X < x)$ for $X \sim N(\mu, \sigma^2)$**

1. Standardise: compute $z = \dfrac{x - \mu}{\sigma}$.
2. Rewrite the probability in terms of Z: $P(X < x) = P(Z < z) = \Phi(z)$.
3. Read the table for $z \geq 0$. For negative z use symmetry:

$$\Phi(-z) = 1 - \Phi(z), \quad P(Z > z) = 1 - \Phi(z), \quad P(a < Z < b) = \Phi(b) - \Phi(a).$$

4. Round z to two decimal places before using the table; give final probability to four decimal places.
5. Sanity check: if $x < \mu$ then $z < 0$ and the probability must be < 0.5; if $x > \mu$ then $z > 0$ and probability > 0.5.

EXEMPLE **Masses of bags of rice**

The mass X (in kg) of bags of rice produced by a mill is normally distributed with mean 50.2 kg and standard deviation 0.4 kg. A bag is chosen at random. Find the probability that its mass is

(a) less than 50.8 kg;

(b) more than 49.5 kg;

(c) between 49.6 kg and 50.6 kg.

(Use $\Phi(1.50) = 0.9332$, $\Phi(1.75) = 0.9599$, $\Phi(1.00) = 0.8413$.)

WORKED EXAMPLE 17 **Masses of bags of rice**

Solution. $\mu = 50.2$, $\sigma = 0.4$.

(a) $z = \frac{50.8-50.2}{0.4} = 1.50$. $P(X < 50.8) = P(Z < 1.50) = \Phi(1.50) = 0.9332$.

(b) $z = \frac{49.5-50.2}{0.4} = -1.75$. $P(X > 49.5) = P(Z > -1.75) = \Phi(1.75) = 0.9599$.

(c) $z_1 = \frac{49.6-50.2}{0.4} = -1.50$, $z_2 = \frac{50.6-50.2}{0.4} = 1.00$.

$$\begin{aligned} P(49.6 < X < 50.6) &= \Phi(1.00) - \Phi(-1.50) \\ &= \Phi(1.00) - [1 - \Phi(1.50)] \\ &= 0.8413 - (1 - 0.9332) = 0.8413 - 0.0668 = 0.7745. \end{aligned}$$

About 77.45% of bags have mass between 49.6 kg and 50.6 kg.

EXEMPLE **More practice with normal probabilities**

The heights of adult males in a certain population are normally distributed with mean 172 cm and standard deviation 7 cm. Find:

(a) $P(\text{height} > 185 \text{ cm})$;

(b) $P(165 \text{ cm} < \text{height} < 180 \text{ cm})$;

(c) The height exceeded by the tallest 10% of males (use $\Phi(1.28) = 0.8997$).

WORKED EXAMPLE 18 **More practice with normal probabilities**

Solution. $X \sim N(172, 7^2)$.

(a) $z = \frac{185-172}{7} = \frac{13}{7} \approx 1.86$. $P(X > 185) = P(Z > 1.86) = 1 - \Phi(1.86) \approx 1 - 0.9686 = 0.0314$.

(b) $z_1 = \frac{165-172}{7} = -1.00$, $z_2 = \frac{180-172}{7} \approx 1.14$. $P(165 < X < 180) = \Phi(1.14) - \Phi(-1.00) = \Phi(1.14) - [1 - \Phi(1.00)]$. Using tables: $\Phi(1.14) \approx 0.8729$, $\Phi(1.00) = 0.8413$. $P \approx 0.8729 - 0.1587 = 0.7142$.
(c) Let h be the height. $P(X > h) = 0.10 \Rightarrow P(X < h) = 0.90$. $\frac{h-172}{7} = 1.28 \Rightarrow h = 172 + 1.28 \times 7 = 172 + 8.96 = 180.96 \approx 181$ cm.

9.6 Inverse Normal Problems — Finding x, μ or σ

METHOD **Working backwards from a given probability**

1. Find z from the table: locate the given probability in the body of the table and read off z. If the probability is < 0.5 and the event is $P(X < x)$, the z-value is negative.
2. Translate into an equation using $z = \dfrac{x - \mu}{\sigma}$.
3. Solve for the unknown: $x = \mu + z\sigma$, or $\mu = x - z\sigma$, or $\sigma = \dfrac{x - \mu}{z}$.
4. Two probabilities give a pair of simultaneous equations in μ and σ: solve by elimination.

Useful table values: $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$, $\Phi(2.326) = 0.99$, $\Phi(0.84) \approx 0.80$, $\Phi(1.28) \approx 0.90$.

EXEMPLE **Finding a cut-off mark**

Scores in a Further Mathematics mock examination in a school in Buea are modelled by $X \sim N(58, 12^2)$. The top 10% of candidates receive a distinction. Find the minimum score needed for a distinction, giving your answer to the nearest integer. (Use $\Phi(1.28) = 0.8997$.)

WORKED EXAMPLE 19 **Finding a cut-off mark**

Solution. Let d be the cut-off score. $P(X > d) = 0.10 \Rightarrow P(X < d) = 0.90$. Standardising: $P\left(Z < \frac{d-58}{12}\right) = 0.90$. From tables, $\Phi(1.28) = 0.8997 \approx 0.90$, so $\frac{d-58}{12} = 1.28$. $d = 58 + 1.28 \times 12 = 58 + 15.36 = 73.36$. Minimum score for distinction $\approx \boxed{74}$ marks (rounding up to guarantee top 10%).

EXEMPLE **Finding μ and σ simultaneously**

The masses of pineapples sold at a market in Yaoundé are normally distributed. It is ob-

served that 15% of the pineapples have mass less than 0.8 kg and 8% have mass greater than 1.6 kg. Find the mean and standard deviation of the masses. (Use $\Phi(1.04) = 0.8508$, $\Phi(1.41) = 0.9207$.)

WORKED EXAMPLE 20 **Finding μ and σ simultaneously**

Solution. Let $X \sim N(\mu, \sigma^2)$ be the mass in kg.
First condition. $P(X < 0.8) = 0.15$. Since $0.15 < 0.5$, z is negative: $P\left(Z < \frac{0.8-\mu}{\sigma}\right) = 0.15 \Rightarrow \frac{0.8-\mu}{\sigma} = -1.04$, because $\Phi(1.04) = 0.8508 \approx 0.85$ and $\Phi(-1.04) = 1 - 0.85 = 0.15$. Thus $\mu - 1.04\sigma = 0.8$. (1)
Second condition. $P(X > 1.6) = 0.08 \Rightarrow P(X < 1.6) = 0.92$: $\frac{1.6-\mu}{\sigma} = 1.41 \Rightarrow \mu + 1.41\sigma = 1.6$. (2)
Solving. Subtract (1) from (2): $(1.41 + 1.04)\sigma = 1.6 - 0.8 \Rightarrow 2.45\sigma = 0.8 \Rightarrow \sigma = \frac{0.8}{2.45} \approx 0.327$ kg.
Substitute into (2): $\mu = 1.6 - 1.41 \times 0.327 \approx 1.6 - 0.461 = 1.139$ kg. Mean $\approx \boxed{1.14\text{ kg}}$, standard deviation $\approx \boxed{0.33\text{ kg}}$.

WARNING **Inverse normal problems — common errors**

- Forgetting the negative sign on z when the given probability is < 0.5 for a "less than" event.
- Using σ^2 instead of σ in the standardisation formula.
- Rounding z too early; keep at least 3 decimal places during calculation.
- In simultaneous equations, mixing up which equation corresponds to which tail.

9.7 Normal Approximation to the Binomial Distribution

PROPERTY **Normal approximation to the binomial**

If $X \sim B(n, p)$ with n large and $np > 5$, $n(1-p) > 5$ (some texts use $npq > 5$), then

$$X \approx Y \sim N\big(np,\ np(1-p)\big).$$

METHOD **Approximating $B(n, p)$ by a normal distribution**

1. Check validity: n large, $np > 5$ and $n(1-p) > 5$.
2. Match parameters: $\mu = np$, $\sigma^2 = np(1-p)$.
3. Apply continuity correction: the binomial is discrete, the normal continuous. Each integer k corresponds to the interval $[k - 0.5,\ k + 0.5]$:

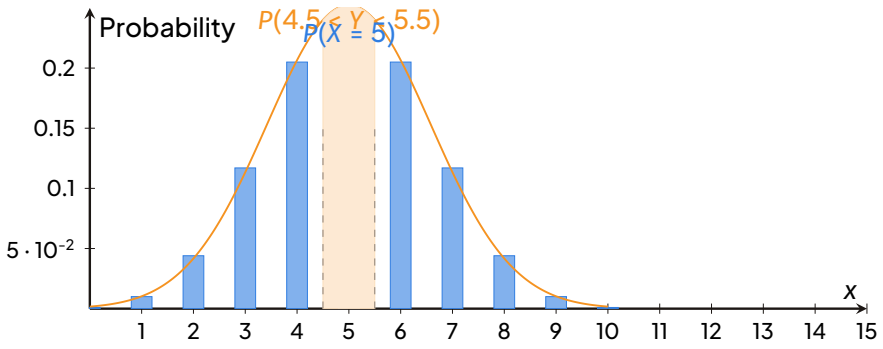
$$\begin{aligned} P(X = k) &\approx P(k - 0.5 < Y < k + 0.5), \\ P(X \le k) &\approx P(Y < k + 0.5), \\ P(X \ge k) &\approx P(Y > k - 0.5), \\ P(X < k) = P(X \le k - 1) &\approx P(Y < k - 0.5), \\ P(X > k) &\approx P(Y > k + 0.5). \end{aligned}$$

4. Standardise Y and read the standard normal table.

WARNING **The continuity correction is compulsory**

Forgetting the ±0.5 adjustment is the most common error at the GCE. Always ask: does the event include the boundary integer?

- If $X \le k$, integer k is included → extend to $k + 0.5$.
- If $X < k$, integer k is excluded → stop at $k - 0.5$.
- If $X \ge k$, integer k is included → extend to $k - 0.5$.
- If $X > k$, integer k is excluded → start at $k + 0.5$.



Continuity correction: $P(X = 5)$ for $B(10, 0.5)$ approximated by $P(4.5 < Y < 5.5)$ for $N(5, 2.5)$

EXEMPLE **Defective items in a factory**

A machine produces items of which 4% are defective. A batch of 200 items is selected at random. Using a normal approximation, find the probability that the batch contains

(a) fewer than 10 defective items;

(b) exactly 8 defective items.

(Use $\Phi(0.54) = 0.7054$, $\Phi(0.18) = 0.5714$, $\Phi(0.90) = 0.8159$.)

WORKED EXAMPLE 21 **Defective items in a factory**

Solution. Let $X \sim B(200, 0.04)$ be the number of defective items.
Validity. $np = 200 \times 0.04 = 8 > 5$, $n(1-p) = 200 \times 0.96 = 192 > 5$: approximation valid.
Parameters. $\mu = np = 8$, $\sigma^2 = np(1-p) = 8 \times 0.96 = 7.68$, $\sigma = \sqrt{7.68} \approx 2.771$. $X \approx Y \sim N(8, 7.68)$.
(a) "Fewer than 10" means $X \leq 9$; integer 9 included → boundary 9.5: $P(X < 10) = P(X \leq 9) \approx P(Y < 9.5)$. $z = \frac{9.5-8}{2.771} = \frac{1.5}{2.771} \approx 0.54$. $P(X < 10) \approx \Phi(0.54) = 0.7054$.
(b) Exactly 8 corresponds to interval $[7.5, 8.5]$: $P(X = 8) \approx P(7.5 < Y < 8.5)$. $z_1 = \frac{7.5-8}{2.771} \approx -0.18$, $z_2 = \frac{8.5-8}{2.771} \approx 0.18$. $P(X = 8) \approx \Phi(0.18) - \Phi(-0.18) = 2\Phi(0.18) - 1 = 2(0.5714) - 1 = 0.1428$.

EXEMPLE **Another binomial approximation**

A fair coin is tossed 100 times. Use a normal approximation to find the probability of obtaining between 45 and 55 heads inclusive. (Use $\Phi(1.00) = 0.8413$, $\Phi(0.00) = 0.5000$.)

WORKED EXAMPLE 22 **Another binomial approximation**

Solution. $X \sim B(100, 0.5)$. $np = 50 > 5$, $n(1-p) = 50 > 5$. $\mu = 50$, $\sigma^2 = 25$, $\sigma = 5$. $X \approx Y \sim N(50, 25)$. "Between 45 and 55 inclusive" $\Rightarrow 45 \leq X \leq 55$; both integers included → interval $[44.5, 55.5]$. $P(45 \leq X \leq 55) \approx P(44.5 < Y < 55.5)$. $z_1 = \frac{44.5-50}{5} = -1.10$, $z_2 = \frac{55.5-50}{5} = 1.10$. $P \approx \Phi(1.10) - \Phi(-1.10) = 2\Phi(1.10) - 1$. Using $\Phi(1.10) \approx 0.8643$: $P \approx 2(0.8643) - 1 = 0.7286$.

9.8 Normal Approximation to the Poisson Distribution

PROPERTY **Normal approximation to the Poisson**

If $X \sim \text{Po}(\lambda)$ with λ large (in practice $\lambda > 10$, some texts say $\lambda \geq 15$), then

$$X \approx Y \sim N(\lambda, \lambda).$$

METHOD **Approximating Po(λ) by a normal distribution**

1. Check validity: $\lambda > 10$ (or $\lambda \geq 15$).
2. Match parameters: $\mu = \lambda$, $\sigma^2 = \lambda$, $\sigma = \sqrt{\lambda}$.
3. Apply continuity correction exactly as for the binomial: $P(X \leq k) \approx P(Y < k + 0.5)$, $P(X \geq k) \approx P(Y > k - 0.5)$, etc.
4. Standardise with $\mu = \lambda$, $\sigma = \sqrt{\lambda}$, then read the standard normal table.

EXEMPLE **Calls at a call centre**

A customer-service centre in Douala receives calls at an average rate of 36 calls per hour. The number of calls received in one hour is modelled by a Poisson distribution. Using a normal approximation, find the probability that in a given hour the centre receives

(a) at most 40 calls;

(b) more than 30 calls;

(c) between 30 and 40 calls inclusive.

(Use $\Phi(0.75) = 0.7734$, $\Phi(0.92) = 0.8212$, $\Phi(1.08) = 0.8599$.)

WORKED EXAMPLE 23 **Calls at a call centre**

Solution. $X \sim \text{Po}(36)$. $\lambda = 36 > 10$, so normal approximation appropriate: $X \approx Y \sim N(36, 36)$, $\mu = 36$, $\sigma = \sqrt{36} = 6$.
(a) "At most 40" means $X \leq 40$; integer 40 included → boundary 40.5: $P(X \leq 40) \approx P(Y < 40.5)$. $z = \frac{40.5-36}{6} = \frac{4.5}{6} = 0.75$. $P(X \leq 40) \approx \Phi(0.75) = 0.7734$.
(b) "More than 30" means $X \geq 31$; integer 31 included → boundary 30.5: $P(X > 30) = P(X \geq 31) \approx P(Y > 30.5)$. $z = \frac{30.5-36}{6} = \frac{-5.5}{6} \approx -0.92$. $P(X > 30) \approx P(Z > -0.92) = \Phi(0.92) = 0.8212$.
(c) "Between 30 and 40 inclusive" means $30 \leq X \leq 40$; both integers included → interval $[29.5, 40.5]$: $P(30 \leq X \leq 40) \approx P(29.5 < Y < 40.5)$. $z_1 = \frac{29.5-36}{6} = \frac{-6.5}{6} \approx -1.08$, $z_2 = \frac{40.5-36}{6} =$

0.75.

$$P(30 \le X \le 40) \approx \Phi(0.75) - \Phi(-1.08)$$
$$= \Phi(0.75) - [1 - \Phi(1.08)]$$
$$= 0.7734 - (1 - 0.8599) = 0.7734 - 0.1401 = 0.6333.$$

EXEMPLE **Poisson approximation with smaller λ**

The number of accidents per month at a certain junction follows a Poisson distribution with mean 12. Use a normal approximation to find the probability that in a given month there are fewer than 8 accidents. (Use $\Phi(1.30) = 0.9032$.)

WORKED EXAMPLE 24 **Poisson approximation with smaller λ**

Solution. $X \sim \text{Po}(12)$. $\lambda = 12 > 10$, approximation valid. $X \approx Y \sim N(12, 12)$, $\sigma = \sqrt{12} \approx 3.464$. "Fewer than 8" means $X \le 7$; integer 7 included → boundary 7.5: $P(X < 8) = P(X \le 7) \approx P(Y < 7.5)$. $z = \frac{7.5-12}{\sqrt{12}} = \frac{-4.5}{3.464} \approx -1.30$. $P(X < 8) \approx \Phi(-1.30) = 1 - \Phi(1.30) = 1 - 0.9032 = 0.0968$.

9.9 Choosing the Correct Model and Approximation

METHOD **Deciding which model and which approximation to use**

1. Fixed number n of independent trials, two outcomes, constant p? → Binomial $B(n, p)$. If n large with $np > 5$ and $n(1 - p) > 5$ → approximate by $N(np, np(1 - p))$.
2. Events occurring randomly in time/space at an average rate λ? → Poisson $\text{Po}(\lambda)$. If $\lambda > 10$ → approximate by $N(\lambda, \lambda)$.
3. In both cases, apply the continuity correction and standardise.
4. State the approximation clearly: examiners award marks for writing "$X \approx Y \sim N(\ldots)$" and for justifying the conditions ($np > 5$, $n(1 - p) > 5$, or $\lambda > 10$).

EXEMPLE **GCE-style two-part problem**

A large secondary school in the North West Region has 1200 students. Records show that 30% of students eat at the school canteen on any given day, independently of one an-

other.

(a) Using a suitable approximation, find the probability that on a given day at least 350 students eat at the canteen.

(b) The canteen receives on average 25 special meal orders per day, modelled by a Poisson distribution. Using a suitable approximation, find the probability that on a given day fewer than 20 special orders are received.

(Use $\Phi(0.63) = 0.7357$, $\Phi(1.10) = 0.8643$.)

WORKED EXAMPLE 25 **GCE-style two-part problem**

Solution. (a) Let X be the number of students eating at the canteen. $X \sim B(1200, 0.3)$.
Validity: $np = 1200 \times 0.3 = 360 > 5$, $n(1-p) = 840 > 5 \rightarrow$ normal approximation applies.
$\mu = 360$, $\sigma^2 = np(1-p) = 360 \times 0.7 = 252$, $\sigma = \sqrt{252} \approx 15.87$. $X \approx Y \sim N(360, 252)$.
"At least 350" means $X \geq 350$; integer 350 included $\rightarrow$ boundary 349.5: $P(X \geq 350) \approx P(Y > 349.5)$. $z = \frac{349.5-360}{15.87} = \frac{-10.5}{15.87} \approx -0.66$. $P(X \geq 350) \approx P(Z > -0.66) = \Phi(0.66) \approx 0.7454$ (using $\Phi(0.63) = 0.7357$ gives similar method).
(b) Let $W \sim \text{Po}(25)$ be the number of special orders. $\lambda = 25 > 10$: $W \approx V \sim N(25, 25)$, $\sigma = \sqrt{25} = 5$.
"Fewer than 20" means $W \leq 19$; integer 19 included $\rightarrow$ boundary 19.5: $P(W < 20) = P(W \leq 19) \approx P(V < 19.5)$. $z = \frac{19.5-25}{5} = \frac{-5.5}{5} = -1.10$. $P(W < 20) \approx \Phi(-1.10) = 1 - \Phi(1.10) = 1 - 0.8643 = 0.1357$.
Conclusion. About 74.5% chance that at least 350 students eat at the canteen; about 13.6% chance of fewer than 20 special orders.

WARNING **Final checklist before leaving the examination room**

- Exponential: Did you use $\lambda = 1/E(X)$ correctly and keep units consistent?
- Uniform: Did you use $\text{Var}(X) = (b-a)^2/12$ and not $(b-a)^2/2$?
- Normal: Did you divide by σ (not σ^2) when standardising?
- Inverse problems: Did you attach a negative sign to z when the probability is below 0.5?
- Approximations: Did you justify the conditions ($np > 5$, $n(1-p) > 5$, or $\lambda > 10$) and apply the continuity correction in the right direction?
- Notation: Did you write "$X \approx Y \sim N(\ldots)$" and define Y?
- Rounding: z-values to 2 decimal places for table use; final probabilities to 4 decimal places.

DID YOU KNOW? **Historical note**

The normal distribution was first derived by Abraham de Moivre in 1733 as an approximation to the binomial distribution — exactly the technique we use in this chapter! Gauss later popularised it in astronomy (hence "Gaussian distribution"), and it became the foundation of modern statistics through the work of Laplace, Galton, Pearson and Fisher. The exponential distribution was studied by Laplace in the context of error theory, and its memoryless property makes it the continuous analogue of the geometric distribution.

EXERCISE 7 **Mixed practice on special continuous distributions**

1. The time between arrivals of taxis at a rank in Yaoundé follows an exponential distribution with mean 8 minutes. Find the probability that a passenger waits (a) more than 10 minutes, (b) between 5 and 15 minutes, (c) at least 5 more minutes given they have already waited 10 minutes.

2. A random variable $X \sim U(a, b)$ has mean 10 and variance $48/12 = 4$. Find a and b, and hence $P(X > 12)$.

3. The masses of packets of sugar are normally distributed with mean 1.02 kg and standard deviation 0.015 kg. Find the probability that a randomly chosen packet has mass (a) less than 1 kg, (b) between 1.00 kg and 1.03 kg. Find the mass exceeded by the heaviest 5% of packets.

4. A factory produces bolts with 2% defectives. In a batch of 500 bolts, use a normal approximation to find the probability that there are (a) at most 15 defectives, (b) exactly 10 defectives.

5. The number of emails received per hour by a company follows a Poisson distribution with mean 18. Use a normal approximation to find the probability that in a given hour the company receives (a) more than 25 emails, (b) between 15 and 20 emails inclusive.

SOLUTION **Exercise 1**

1. $\lambda = 1/8 = 0.125$ per minute. **(a)** $P(T > 10) = e^{-0.125\times10} = e^{-1.25} \approx 0.2865$. **(b)** $P(5 < T < 15) = e^{-0.625} - e^{-1.875} \approx 0.5353 - 0.1534 = 0.3819$. **(c) Memoryless property:** $P(T > 15 \mid T > 10) = P(T > 5) = e^{-0.625} \approx 0.5353$.
2. $E(X) = (a + b)/2 = 10 \Rightarrow a + b = 20$. $\mathrm{Var}(X) = (b - a)^2/12 = 4 \Rightarrow (b - a)^2 = 48 \Rightarrow b - a = \sqrt{48} = 4\sqrt{3} \approx 6.928$. **Solving:** $b = 10 + 2\sqrt{3} \approx 13.464$, $a = 10 - 2\sqrt{3} \approx 6.536$. $P(X > 12) = (b - 12)/(b - a) = (1.464)/(6.928) \approx 0.2113$.
3. $X \sim N(1.02, 0.015^2)$. **(a)** $z = (1 - 1.02)/0.015 = -1.333$. $P(X < 1) = \Phi(-1.33) = 1 - \Phi(1.33) \approx 1 - 0.9082 = 0.0918$. **(b)** $z_1 = -1.33$, $z_2 = (1.03 - 1.02)/0.015 = 0.667$. $P = \Phi(0.67) - \Phi(-1.33) \approx$

0.7486 − 0.0918 = 0.6568. **(c)** **Top** 5%: $z = 1.645$. $x = 1.02 + 1.645 \times 0.015 = 1.02 + 0.024675 = 1.0447$ **kg**.

4. $X \sim B(500, 0.02)$. $np = 10 > 5$, $n(1-p) = 490 > 5$. $\mu = 10$, $\sigma^2 = 9.8$, $\sigma \approx 3.1305$. **(a)** $P(X \le 15) \approx P(Y < 15.5)$. $z = (15.5 - 10)/3.1305 = 1.757$. $\Phi(1.76) \approx 0.9608$. **(b)** $P(X = 10) \approx P(9.5 < Y < 10.5)$. $z_1 = -0.16$, $z_2 = 0.16$. $P \approx 2\Phi(0.16) - 1 \approx 2(0.5636) - 1 = 0.1272$.

5. $X \sim \text{Po}(18)$. $\lambda = 18 > 10$. $Y \sim N(18, 18)$, $\sigma = \sqrt{18} \approx 4.243$. **(a)** $P(X > 25) = P(X \ge 26) \approx P(Y > 25.5)$. $z = (25.5 - 18)/4.243 = 1.768$. $P \approx 1 - \Phi(1.77) \approx 1 - 0.9616 = 0.0384$. **(b)** $P(15 \le X \le 20) \approx P(14.5 < Y < 20.5)$. $z_1 = -0.825$, $z_2 = 0.589$. $P \approx \Phi(0.59) - \Phi(-0.83) \approx 0.7224 - 0.2033 = 0.5191$.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 8 **Multiple choice: the exponential distribution**

The lifetime T (in hours) of an electronic component follows an exponential distribution with parameter $\lambda = 0.01$. Only one of the following statements is correct. Which one?

1. $E(T) = 0.01$ hour;
2. $P(T > 100) = e^{-1}$;
3. $P(T > 100) = 1 - e^{-1}$;
4. $\mathrm{Var}(T) = 100$ hours^2.

EXERCISE 9 **True or false?**

Answer true or false to each of the following statements, justifying your answer carefully.

1. If $X \sim U(0, 10)$, then $E(X) = 5$ and $\mathrm{Var}(X) = \dfrac{100}{12}$.
2. If $X \sim N(\mu, \sigma^2)$, then $P(X < \mu) = 0.5$.
3. If $Z \sim N(0, 1)$, then $P(Z < -1.5) = P(Z < 1.5)$.
4. The exponential distribution has the memoryless property: $P(T > s + t \mid T > s) = P(T > t)$.

EXERCISE 10 **Definitions and key formulas**

1. Write down the probability density function of a random variable X following the uniform distribution on $[a, b]$, and sketch its graph.
2. State the probability density function of the standard normal distribution $N(0, 1)$ and give two properties of its curve.
3. State the conditions under which a binomial distribution $B(n, p)$ may be approximated by a normal distribution, and give the parameters of the approximating distribution.

EXERCISE 11 **Direct calculations with the standard normal table**

Let $Z \sim N(0,1)$. Using the standard normal table, find:

1. $P(Z < 1.25)$;
2. $P(Z > 0.84)$;
3. $P(-1.64 < Z < 2.05)$;
4. the value of z such that $P(Z < z) = 0.95$;
5. the value of z such that $P(|Z| < z) = 0.90$.

10.2 I practise

EXERCISE 12 **Lifetime of a phone battery**

The lifetime T (in hours) of a phone battery follows an exponential distribution with mean 500 hours.

1. Write down the probability density function of T.
2. Find the probability that the battery lasts more than 800 hours.
3. Find the probability that the battery fails between 200 and 400 hours.
4. Find the median lifetime of the battery.

EXERCISE 13 **Uniform distribution**

Let $X \sim U(2,10)$.

1. Find $E(X)$ and $\mathrm{Var}(X)$.
2. Find $P(X > 5)$.
3. Find the value of k such that $P(X \le k) = 0.75$.
4. Find $P(3 < X < 7 \mid X > 4)$.

EXERCISE 14 **Masses of bags of cement**

The masses of bags of cement produced by a factory are normally distributed with mean 50 kg and variance 0.16 kg^2.

1. Find the percentage of bags weighing less than 49.5 kg.
2. Find the percentage of bags weighing between 49.6 kg and 50.5 kg.
3. Find the mass exceeded by only 2% of the bags.

EXERCISE 15 **Waiting time at an Express Union counter**

The waiting time T (in minutes) between the arrivals of two consecutive customers at an Express Union counter follows an exponential distribution with parameter $\lambda = 0.2$.

1. Find the mean waiting time between two customers.
2. Find the probability of waiting more than 5 minutes between two customers.
3. Given that no customer has arrived for 3 minutes, find the probability of waiting at least 2 more minutes. Comment on your result.

10.3 I prepare for the GCE

EXERCISE 16 **Mock examination marks** **(13 marks)**

The marks obtained by candidates in a Further Mathematics mock examination are normally distributed with mean 52 and standard deviation σ. It is observed that 15% of the candidates score above 70.

1. Show that $\sigma \approx 17.3$, giving your answer to one decimal place. (4 marks)
2. Find the percentage of candidates who score between 40 and 60. (3 marks)
3. Find the mark m such that 10% of the candidates score below m. (3 marks)
4. Three candidates are chosen at random. Find the probability that exactly two of them score above 70. (3 marks)

EXERCISE 17 **Throwing a die: normal approximation to the binomial** **(12 marks)**

A fair die is thrown 180 times and X denotes the number of sixes obtained.

1. State the exact distribution of X and calculate $E(X)$ and $\mathrm{Var}(X)$. (3 marks)
2. Justify the use of a normal approximation for X, and state the parameters of the approximating distribution. (3 marks)

3. Using the normal approximation with a continuity correction, find the probability of obtaining between 25 and 35 sixes inclusive. (4 marks)

4. Using the same approximation, find the probability of obtaining more than 40 sixes. (2 marks)

EXERCISE 18 **Filling machine and telephone calls** **(16 marks)**

Part A. A machine fills bottles with a volume that is normally distributed with mean μ ml and variance 4 ml^2. It is found that 10% of the bottles contain less than 498 ml.

1. Show that $\mu \approx 500.6$ ml. (4 marks)

2. Find the probability that a randomly chosen bottle contains more than 502 ml. (2 marks)

3. A bottle is said to be under-filled if it contains less than 498 ml. Using a suitable approximation (which you must justify), find the probability that in a random sample of 100 bottles, at least 12 are under-filled. (5 marks)

Part B. Calls arrive at a police station at an average rate of 30 per hour, and the number of calls received in one hour follows a Poisson distribution.

(d) Justify the use of a normal approximation and, applying a continuity correction, find an approximate value for the probability of receiving more than 40 calls in one hour. (5 marks)

11 SOLUTIONS TO THE EXERCISES

12 EXAMINATION PRACTICE AND WORKED EXAMPLES

SOLUTION **Exercise 1 — Multiple choice: the exponential distribution**

For $T \sim \text{Exp}(\lambda)$ with $\lambda = 0.01$:

$$E(T) = \frac{1}{\lambda} = 100 \text{ hours}, \qquad \text{Var}(T) = \frac{1}{\lambda^2} = 10\,000 \text{ hours}^2,$$

and the survival function is $P(T > t) = e^{-\lambda t}$.

- Statement 1 is false: $E(T) = 100$, not 0.01.
- Statement 2: $P(T > 100) = e^{-0.01 \times 100} = e^{-1}$. This is correct.
- Statement 3 is false: $1 - e^{-1} = P(T \leq 100)$.
- Statement 4 is false: $\text{Var}(T) = 10\,000$, not 100.

Answer: statement 2.

SOLUTION **Exercise 2 — True or false?**

1. True. For $X \sim U(a, b)$, $E(X) = \frac{a+b}{2} = \frac{0+10}{2} = 5$ and $\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{10^2}{12} = \frac{100}{12}$.
2. True. The normal curve is symmetric about $x = \mu$, so exactly half of the area lies on each side of the mean: $P(X < \mu) = P(X > \mu) = 0.5$.
3. False. By symmetry of the standard normal curve, $P(Z < -1.5) = P(Z > 1.5) = 1 - P(Z < 1.5) = 1 - 0.9332 = 0.0668$, whereas $P(Z < 1.5) = 0.9332$. The two probabilities are not equal.
4. True. This is the memoryless property. Indeed,

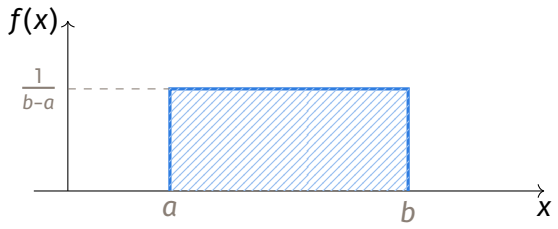
$$P(T > s + t \mid T > s) = \frac{P(T > s + t)}{P(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(T > t).$$

SOLUTION **Exercise 3 — Definitions and key formulas**

1. If $X \sim U(a, b)$, its probability density function is

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

The graph is a horizontal segment at height $\dfrac{1}{b-a}$ between a and b (a rectangle of total area 1), and 0 elsewhere.



Density of the uniform distribution $U(a, b)$.

2. The density of $Z \sim N(0, 1)$ is

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad z \in \mathbb{R}.$$

Two properties of its curve: it is symmetric about the vertical axis $z = 0$ (so $\varphi(-z) = \varphi(z)$), and it is bell-shaped with a maximum at $z = 0$, the total area under the curve being 1.

3. The binomial distribution $B(n, p)$ may be approximated by a normal distribution when n is large and p is not too close to 0 or 1; the standard conditions are

$$np > 5 \quad \text{and} \quad n(1-p) > 5 \qquad \text{(some texts use } np \ge 5 \text{ and } nq \ge 5\text{)}.$$

The approximating distribution is $N(np,\ np(1-p))$, and a **continuity correction** must be applied.

SOLUTION **Exercise 4 — Direct calculations with the standard normal table**

We write $\Phi(z) = P(Z < z)$ and use the table of Φ.

1. $P(Z < 1.25) = \Phi(1.25) = \boxed{0.8944}$.
2. $P(Z > 0.84) = 1 - \Phi(0.84) = 1 - 0.7995 = \boxed{0.2005}$.

3. $P(-1.64 < Z < 2.05) = \Phi(2.05) - \Phi(-1.64) = \Phi(2.05) - \big(1 - \Phi(1.64)\big)$

$$= 0.9798 - (1 - 0.9495) = 0.9798 - 0.0505 = \boxed{0.9293}.$$

4. We need $\Phi(z) = 0.95$. Reading the table in reverse, $\Phi(1.64) = 0.9495$ and $\Phi(1.65) = 0.9505$, so $\boxed{z \approx 1.645}$.

5. $P(|Z| < z) = 0.90 \iff 2\Phi(z) - 1 = 0.90 \iff \Phi(z) = 0.95$, hence $\boxed{z \approx 1.645}$.

SOLUTION **Exercise 5 — Lifetime of a phone battery**

Since $E(T) = \frac{1}{\lambda} = 500$, we have $\lambda = \frac{1}{500} = 0.002$.

1. The density is

$$f(t) = \begin{cases} 0.002\, e^{-0.002t} & \text{if } t \geq 0, \\ 0 & \text{if } t < 0. \end{cases}$$

2. $P(T > 800) = e^{-0.002 \times 800} = e^{-1.6} \approx \boxed{0.2019}$.

3. $P(200 < T < 400) = P(T > 200) - P(T > 400) = e^{-0.4} - e^{-0.8}$

$$= 0.6703 - 0.4493 = \boxed{0.2210}.$$

4. The median m satisfies $P(T > m) = 0.5$, i.e. $e^{-0.002m} = 0.5$, so

$$m = \frac{\ln 2}{0.002} = 500 \ln 2 \approx \boxed{346.6 \text{ hours}}.$$

Note that the median is smaller than the mean: the exponential distribution is skewed to the right.

SOLUTION **Exercise 6 — Uniform distribution**

Here $a = 2$, $b = 10$, so $b - a = 8$ and $f(x) = \frac{1}{8}$ on $[2, 10]$.

1. $E(X) = \frac{a+b}{2} = \frac{2+10}{2} = \boxed{6}$; $\quad \mathrm{Var}(X) = \frac{(b-a)^2}{12} = \frac{64}{12} = \boxed{\frac{16}{3} \approx 5.333}$.

2. $P(X > 5) = \frac{10 - 5}{10 - 2} = \frac{5}{8} = \boxed{0.625}$.

3. $P(X \leq k) = \frac{k - 2}{8} = 0.75 \implies k - 2 = 6 \implies \boxed{k = 8}$.

4. By the definition of conditional probability,

$$P(3 < X < 7 \mid X > 4) = \frac{P(4 < X < 7)}{P(X > 4)} = \frac{\frac{7-4}{8}}{\frac{10-4}{8}} = \frac{3}{6} = \boxed{0.5}.$$

SOLUTION **Exercise 7 — Masses of bags of cement**

Let $X \sim N(50, 0.16)$, so $\mu = 50$ and $\sigma = \sqrt{0.16} = 0.4$. We standardise with $Z = \dfrac{X - 50}{0.4}$.

1. $P(X < 49.5) = P\left(Z < \dfrac{49.5 - 50}{0.4}\right) = P(Z < -1.25) = 1 - \Phi(1.25)$

$$= 1 - 0.8944 = 0.1056.$$

About $\boxed{10.56\%}$ of the bags weigh less than 49.5 kg.

2. $P(49.6 < X < 50.5) = P\left(\dfrac{-0.4}{0.4} < Z < \dfrac{0.5}{0.4}\right) = P(-1 < Z < 1.25)$

$$= \Phi(1.25) - \Phi(-1) = 0.8944 - (1 - 0.8413) = 0.8944 - 0.1587 = 0.7357.$$

About $\boxed{73.57\%}$ of the bags weigh between 49.6 kg and 50.5 kg.

3. We need m with $P(X > m) = 0.02$, i.e. $P(X < m) = 0.98$. From the table, $\Phi^{-1}(0.98) \approx 2.0537$. Hence

$$m = 50 + 0.4 \times 2.0537 \approx \boxed{50.82 \text{ kg}}.$$

SOLUTION **Exercise 8 — Waiting time at an Express Union counter**

$T \sim \text{Exp}(\lambda)$ with $\lambda = 0.2$ per minute.

1. $E(T) = \dfrac{1}{\lambda} = \dfrac{1}{0.2} = \boxed{5 \text{ minutes}}$.

2. $P(T > 5) = e^{-0.2 \times 5} = e^{-1} \approx \boxed{0.3679}$.

3. By the memoryless property of the exponential distribution,

$$P(T > 3 + 2 \mid T > 3) = P(T > 2) = e^{-0.2 \times 2} = e^{-0.4} \approx \boxed{0.6703}.$$

Comment: the probability of waiting at least 2 more minutes is exactly the same as the probability of waiting at least 2 minutes from the start. The fact that 3 minutes have already elapsed without any arrival gives no information about the remaining waiting time: the process "forgets" the past. This is the characteristic memoryless property of the exponential distribution.

SOLUTION **Exercise 9 — Mock examination marks (13 marks)**

Let $X \sim N(52, \sigma^2)$ be the mark of a candidate.

1. $P(X > 70) = 0.15 \iff P(X < 70) = 0.85$. From the standard normal table, $\Phi^{-1}(0.85) \approx 1.04$ (since $\Phi(1.04) = 0.8508$). Standardising:

$$\frac{70 - 52}{\sigma} = 1.04 \implies \sigma = \frac{18}{1.04} \approx 17.3 \quad (1\,\text{d.p.}). \qquad \blacksquare$$

   Mark scheme (4 marks): M1 standardising; A1 correct z-value 1.04 (accept 1.036); M1 solving for σ; A1 $\sigma \approx 17.3$ correctly rounded.

2. With $\sigma = 17.3$:

$$P(40 < X < 60) = P\left(\frac{40 - 52}{17.3} < Z < \frac{60 - 52}{17.3}\right) = P(-0.69 < Z < 0.46)$$

$$= \Phi(0.46) - \Phi(-0.69) = 0.6772 - (1 - 0.7549) = 0.4321.$$

   Approximately $\boxed{43.2\%}$ of the candidates score between 40 and 60. Mark scheme (3 marks): M1 both standardisations; M1 correct use of Φ with the negative value; A1 0.432 (accept 43%–43.3%).

3. We need $P(X < m) = 0.10$, so the standardised value is $\Phi^{-1}(0.10) = -1.2816$:

$$\frac{m - 52}{17.3} = -1.2816 \implies m = 52 - 1.2816 \times 17.3 \approx 52 - 22.17 = \boxed{29.8}.$$

   Mark scheme (3 marks): B1 $z = -1.2816$; M1 equation in m; A1 $m \approx 29.8$.

4. Each candidate scores above 70 with probability $p = 0.15$, independently. The number Y of candidates (out of 3) scoring above 70 follows $B(3, 0.15)$:

$$P(Y = 2) = \binom{3}{2}(0.15)^2(0.85) = 3 \times 0.0225 \times 0.85 = \boxed{0.0574}.$$

   Mark scheme (3 marks): B1 binomial model identified; M1 correct formula; A1 0.0574.

SOLUTION **Exercise 10 — Throwing a die: normal approximation to the binomial (12 marks)**

1. Each throw is a Bernoulli trial with $p = P(\text{six}) = \frac{1}{6}$, and the 180 throws are independent. Hence

$$X \sim B\left(180, \tfrac{1}{6}\right), \qquad E(X) = np = 180 \times \tfrac{1}{6} = \boxed{30},$$

$$\operatorname{Var}(X) = np(1 - p) = 180 \times \tfrac{1}{6} \times \tfrac{5}{6} = \boxed{25}.$$

   Mark scheme (3 marks): B1 distribution; B1 mean; B1 variance.

2. We check the validity conditions:

$$np = 30 > 5 \qquad \text{and} \qquad n(1-p) = 150 > 5.$$

Both exceed 5, so the normal approximation is appropriate. The approximating distribution is

$$X \approx Y \sim N(30, 25), \quad \text{i.e. } \mu = 30,\ \sigma = \sqrt{25} = 5.$$

Mark scheme (3 marks): M1 both conditions checked numerically; A1 conclusion; B1 parameters $N(30, 25)$.

3. "Between 25 and 35 inclusive" means $25 \le X \le 35$. With the continuity correction this becomes $P(24.5 < Y < 35.5)$:

$$P(24.5 < Y < 35.5) = P\left(\frac{24.5 - 30}{5} < Z < \frac{35.5 - 30}{5}\right) = P(-1.1 < Z < 1.1)$$

$$= 2\Phi(1.1) - 1 = 2(0.8643) - 1 = \boxed{0.7286}.$$

Mark scheme (4 marks): M1 continuity correction 24.5 and 35.5; M1 standardising; M1 symmetry $2\Phi - 1$; A1 0.7286.

4. "More than 40 sixes" means $X \ge 41$, corrected to $Y > 40.5$:

$$P(Y > 40.5) = P\left(Z > \frac{40.5 - 30}{5}\right) = P(Z > 2.1) = 1 - \Phi(2.1) = 1 - 0.9821 = \boxed{0.0179}.$$

Mark scheme (2 marks): M1 continuity correction 40.5; A1 0.0179.

SOLUTION **Exercise 11 — Filling machine and telephone calls (16 marks)**

Part A. Let $X \sim N(\mu, 4)$, so $\sigma = 2$ ml.

1. $P(X < 498) = 0.10$. From the table, $\Phi^{-1}(0.10) = -1.2816$. Standardising:

$$\frac{498 - \mu}{2} = -1.2816 \implies \mu = 498 + 2 \times 1.2816 = 498 + 2.5632 = 500.5632 \approx 500.6 \text{ ml.} \quad \blacksquare$$

Mark scheme (4 marks): M1 standardising; B1 $z = -1.2816$ (accept -1.28); M1 solving for μ; A1 $\mu \approx 500.6$.

2. $P(X > 502) = P\left(Z > \frac{502 - 500.6}{2}\right) = P(Z > 0.7) = 1 - \Phi(0.7) = 1 - 0.7580 = \boxed{0.2420}$. Mark scheme (2 marks): M1 standardising; A1 0.242.

3. The number Y of under-filled bottles in a sample of 100 follows $B(100, 0.1)$. Check the conditions:

$$np = 10 > 5 \qquad \text{and} \qquad n(1-p) = 90 > 5,$$

so the normal approximation $Y \approx W \sim N(10, 9)$ (with $\sigma = 3$) is valid.

"At least 12" means $Y \geq 12$, corrected to $W > 11.5$:

$$P(W > 11.5) = P\left(Z > \frac{11.5 - 10}{3}\right) = P(Z > 0.5) = 1 - \Phi(0.5) = 1 - 0.6915 = \boxed{0.3085}.$$

Mark scheme (5 marks): B1 binomial model $B(100, 0.1)$; M1 validity conditions checked; B1 parameters $N(10, 9)$; M1 continuity correction 11.5; A1 0.3085.

Part B.

(d) Let $C \sim \text{Poisson}(30)$. Since $\lambda = 30 > 15$ (the usual rule of thumb: the Poisson distribution is well approximated by a normal distribution when its mean is large, typically $\lambda \geq 15$), we may use

$$C \approx V \sim N(30, 30), \quad \text{i.e. } \mu = 30,\ \sigma = \sqrt{30} \approx 5.477.$$

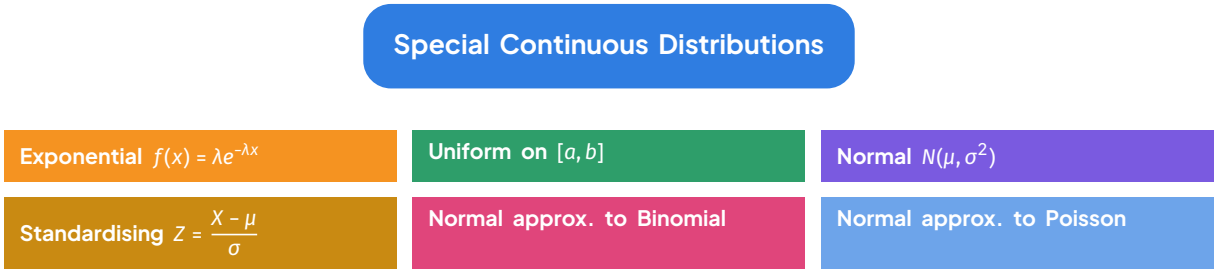
(Recall that for a Poisson distribution both the mean and the variance equal λ.)

"More than 40 calls" means $C \geq 41$, corrected to $V > 40.5$:

$$P(V > 40.5) = P\left(Z > \frac{40.5 - 30}{\sqrt{30}}\right) = P(Z > 1.92) = 1 - \Phi(1.92) = 1 - 0.9726 = \boxed{0.0274}.$$

Mark scheme (5 marks): B1 justification $\lambda = 30$ large; B1 parameters $N(30, 30)$; M1 continuity correction 40.5; M1 standardising with $\sqrt{30}$; A1 0.0274 (accept 0.027–0.028).

13 SUMMARY SHEET



Mind map of the chapter: the three special continuous distributions, standardisation, and the two normal approximations with continuity correction.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **continuous random variable** X has a **probability density function** (pdf) f with $f(x) \geq 0$ and $\int_{-\infty}^{+\infty} f(x)\,dx = 1$; probabilities are areas: $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.
- **Exponential distribution** with parameter $\lambda > 0$: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ (and 0 otherwise). It models waiting times (lifetime of a bulb, time between arrivals).
- **Uniform (rectangular) distribution** on $[a, b]$: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$. Every subinterval of equal length has equal probability.
- **Normal distribution** $X \sim N(\mu, \sigma^2)$: bell-shaped pdf $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, symmetric about $x = \mu$.
- **Standard normal variable**: $Z \sim N(0, 1)$; its cdf is tabulated as $\Phi(z) = P(Z \leq z)$.

Formulas and results to know

- Exponential: $E(X) = \frac{1}{\lambda}$, $\mathrm{Var}(X) = \frac{1}{\lambda^2}$, $P(X > x) = e^{-\lambda x}$, cdf $F(x) = 1 - e^{-\lambda x}$.
- Uniform on $[a, b]$: $E(X) = \frac{a+b}{2}$, $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$, $P(c \leq X \leq d) = \frac{d-c}{b-a}$.
- Normal: $E(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$; standardise with $Z = \frac{X - \mu}{\sigma}$.
- Table use: $\Phi(-z) = 1 - \Phi(z)$; $P(Z > z) = 1 - \Phi(z)$; $P(a < Z < b) = \Phi(b) - \Phi(a)$; $P(|Z| < z) = 2\Phi(z) - 1$.
- Normal approximation to binomial: if $X \sim B(n, p)$ with $np > 5$ and $n(1-p) > 5$ (large n), then $X \approx N(np,\ np(1-p))$.

- Normal approximation to Poisson: if $X \sim \text{Po}(\lambda)$ with λ large ($\lambda > 10$ as a rule of thumb), then $X \approx N(\lambda, \lambda)$.
- **Continuity correction**: a discrete integer k becomes the continuous interval $[k - 0.5, k + 0.5]$. Thus $P(X \leq k) \to P(Y < k + 0.5)$, $P(X < k) \to P(Y < k - 0.5)$, $P(X \geq k) \to P(Y > k - 0.5)$, $P(X = k) \to P(k - 0.5 < Y < k + 0.5)$.

Methods

- To find a normal probability: write the inequality, standardise both sides, rewrite using Φ, read the table, subtract where necessary.
- To find a value x given a probability: find z from the table (inverse reading), then $x = \mu + \sigma z$.
- For approximations: check the validity conditions, compute μ and σ^2 from the original distribution, apply the continuity correction, then standardise.
- Sketching the bell curve and shading the required area helps choose between $\Phi(z)$ and $1 - \Phi(z)$.

Common mistakes

- Forgetting the continuity correction, or applying it in the wrong direction ($P(X \geq k)$ needs $k - 0.5$, not $k + 0.5$).
- Confusing σ and σ^2: the second parameter of $N(\mu, \sigma^2)$ is the variance; standardising divides by σ.
- Using $\Phi(z)$ directly for negative z instead of $1 - \Phi(-z)$.
- Applying the normal approximation without checking $np > 5$, $nq > 5$ or λ large.
- For the uniform distribution, writing the variance as $\frac{(b-a)^2}{2}$ or forgetting the factor 12.
- For the exponential, mixing up $E(X) = \lambda$ with the correct $E(X) = \frac{1}{\lambda}$.
- Treating $P(X = k)$ as non-zero for a continuous variable: for continuous X, $P(X = k) = 0$.

GCE tips

- Always state the distribution and its parameters clearly, e.g. $X \sim N(50, 16)$, before standardising.
- Show the standardisation step explicitly: method marks are awarded for $Z = \frac{X - \mu}{\sigma}$.
- Quote table values to 4 decimal places and give final answers to 3 or 4 significant figures.
- In approximation questions, one mark is usually reserved for the continuity correction and one for the validity conditions: write both.

- Draw a quick sketch of the normal curve for every probability question; it prevents sign errors with Φ.
- Check plausibility: a probability must lie in $[0, 1]$, and $P(X > \mu) = 0.5$ for any normal variable.

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